

(12) **United States Patent**
Wada et al.

(10) **Patent No.:** **US 12,410,027 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **PRINTING APPARATUS AND ROLL
HOLDING DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 116 days.

(21) Appl. No.: **18/178,210**

(22) Filed: **Mar. 3, 2023**

(65) **Prior Publication Data**

US 2023/0294944 A1 Sep. 21, 2023

(30) **Foreign Application Priority Data**

Mar. 18, 2022 (JP) 2022-043691

(51) **Int. Cl.**
B65H 16/00 (2006.01)
B65H 16/02 (2006.01)

(52) **U.S. Cl.**
CPC **B65H 16/028** (2013.01); **B65H 16/005**
(2013.01); **B65H 2301/4432** (2013.01); **B65H**
2402/31 (2013.01); **B65H 2402/53** (2013.01);
B65H 2403/51 (2013.01)

(58) **Field of Classification Search**
CPC B65H 18/04; B65H 2301/41335
See application file for complete search history.

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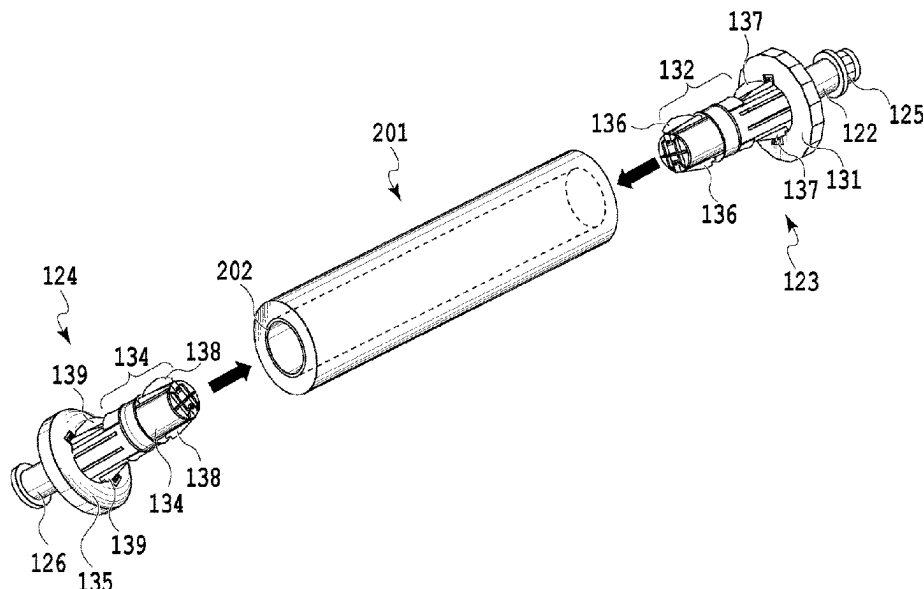
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(57) **ABSTRACT**

Provided is a printing apparatus for performing printing on a sheet drawn out from a roll including: a roll holding unit configured to hold the roll; a support unit configured to rotatably support the roll holding unit; a core portion provided to the roll holding unit and configured to be inserted into a hollow portion of a core pipe of the roll from an end portion of the core pipe to extend along an axis of the core pipe; a contact member provided to the core portion and configured to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and a cam member disposed inside the core portion and configured to pivot to a first position where the contact member is pressed onto the inner wall and to a second position where the contact member is retracted from the inner wall.

11 Claims, 20 Drawing Sheets



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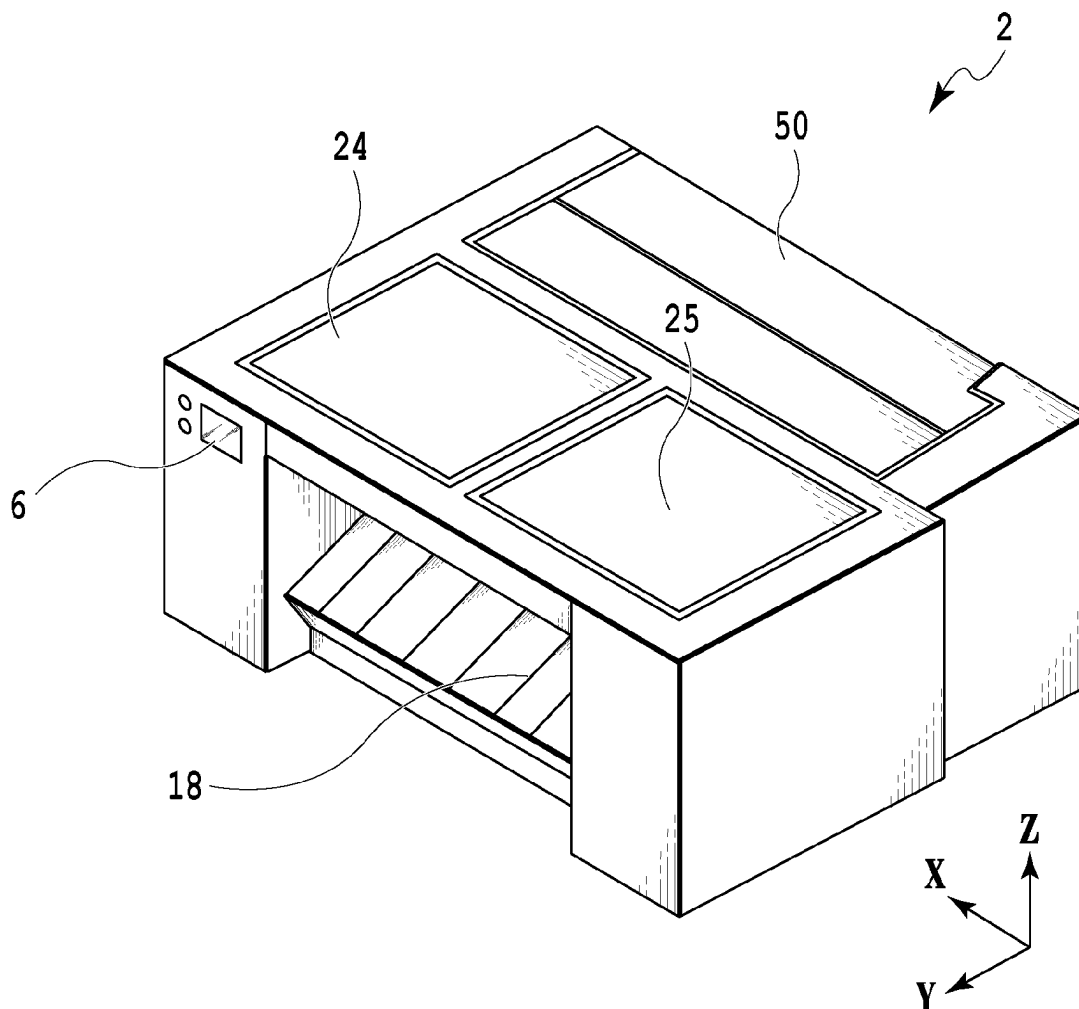


FIG.1



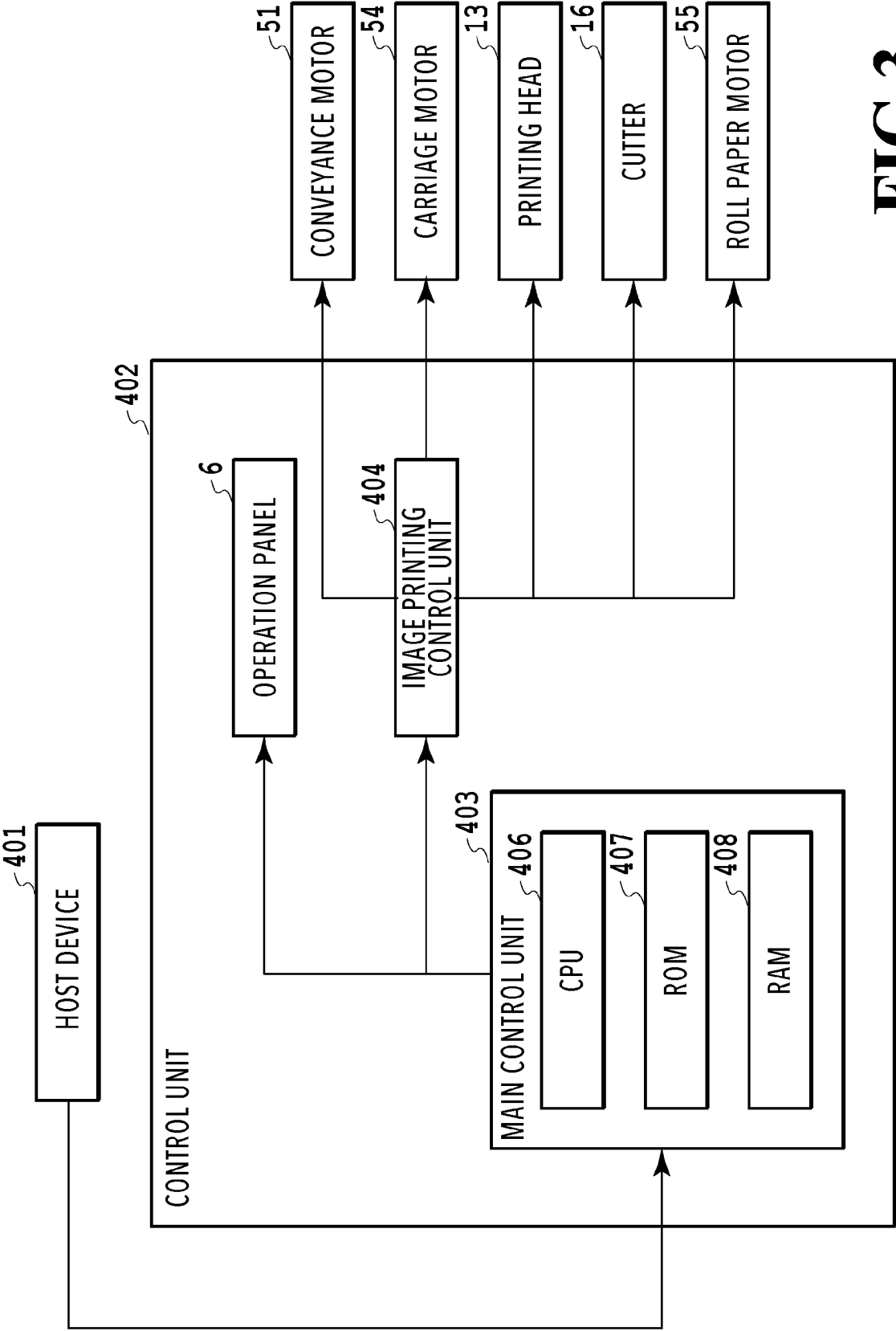


FIG.3

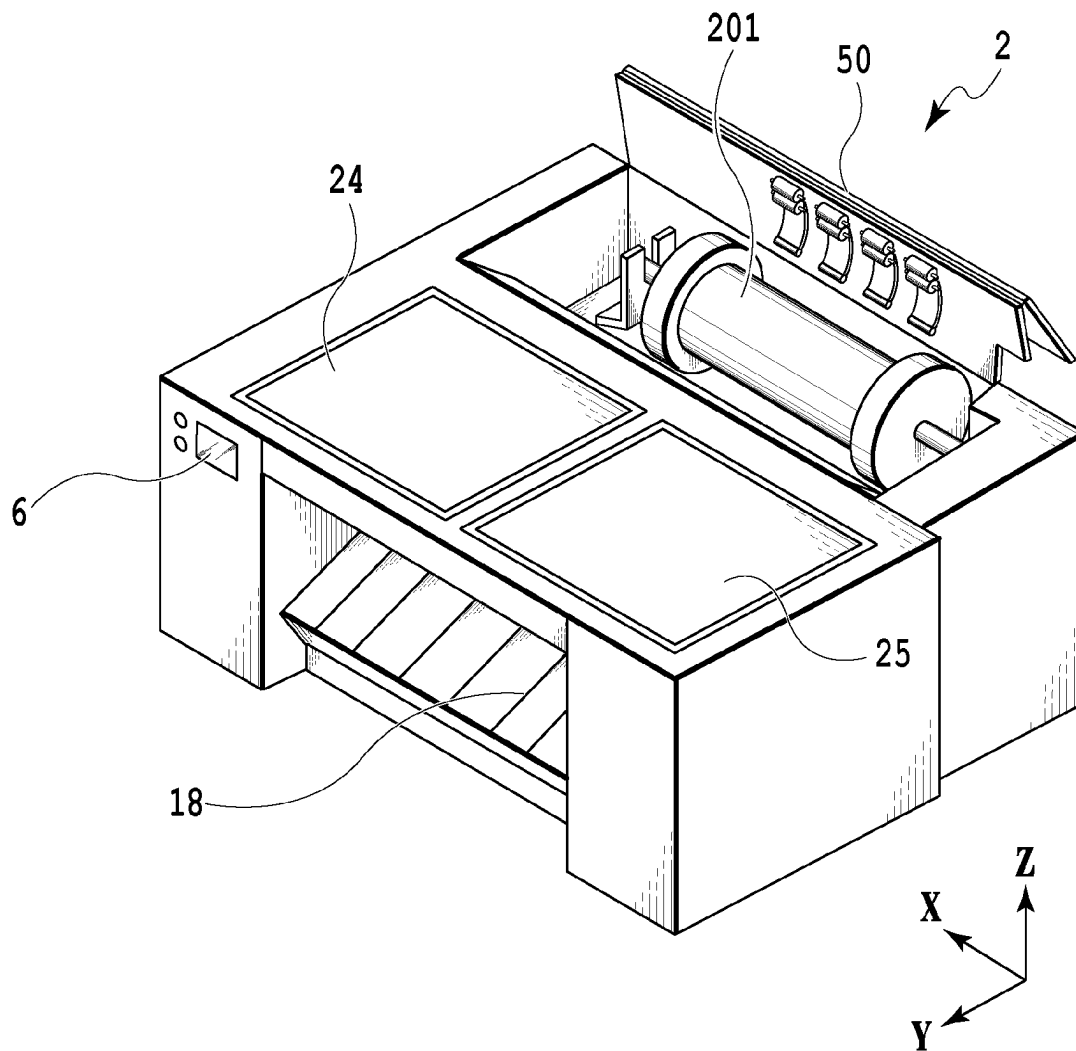
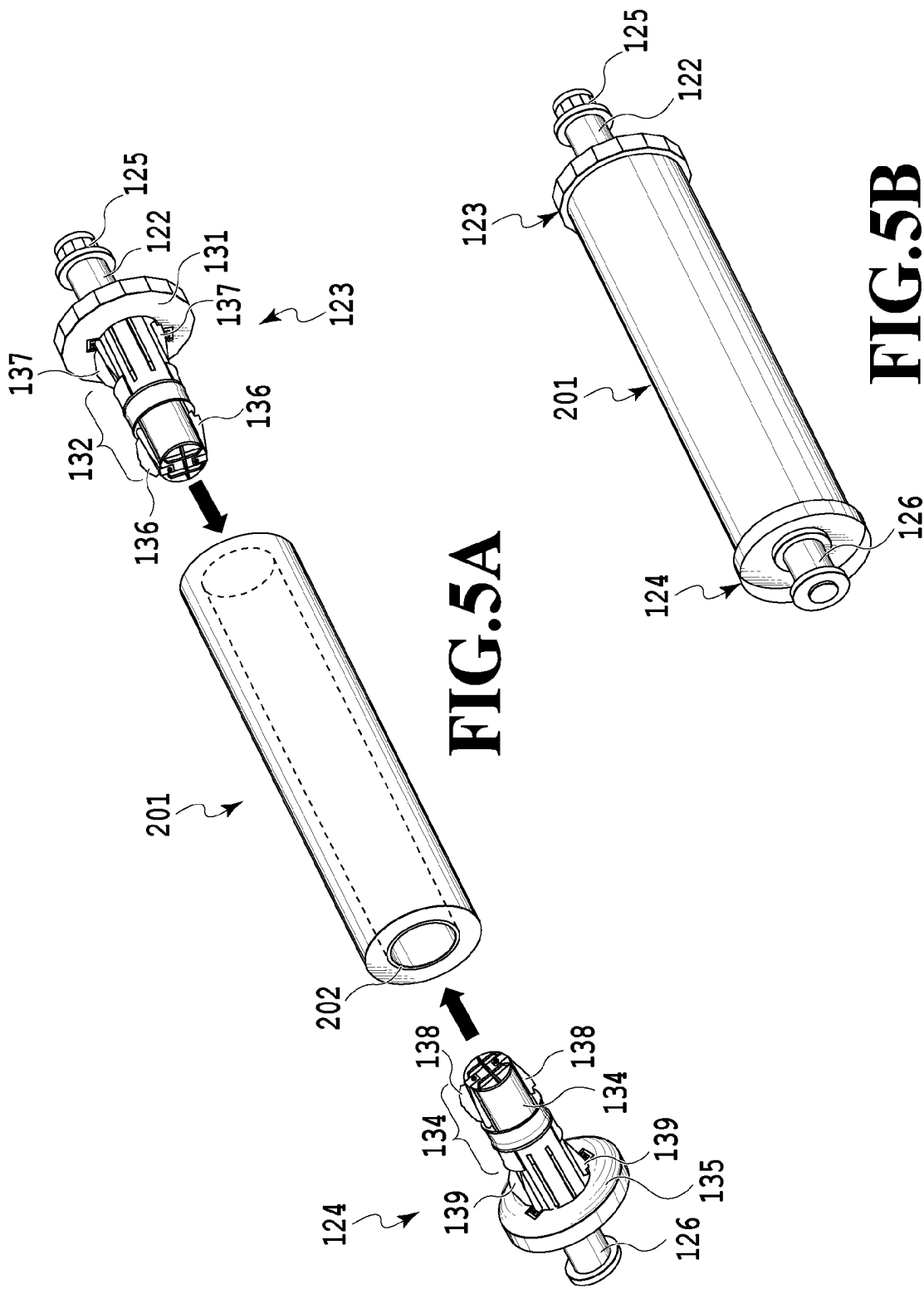


FIG.4



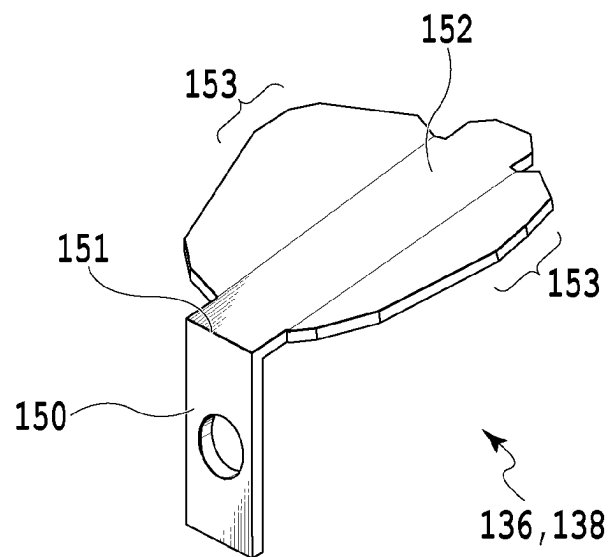


FIG.6A

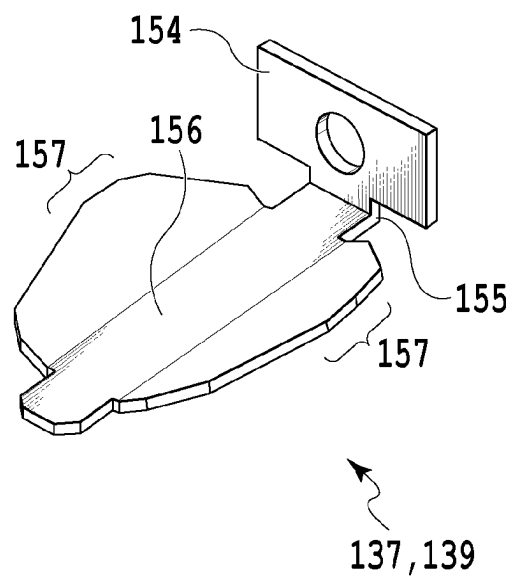


FIG.6B

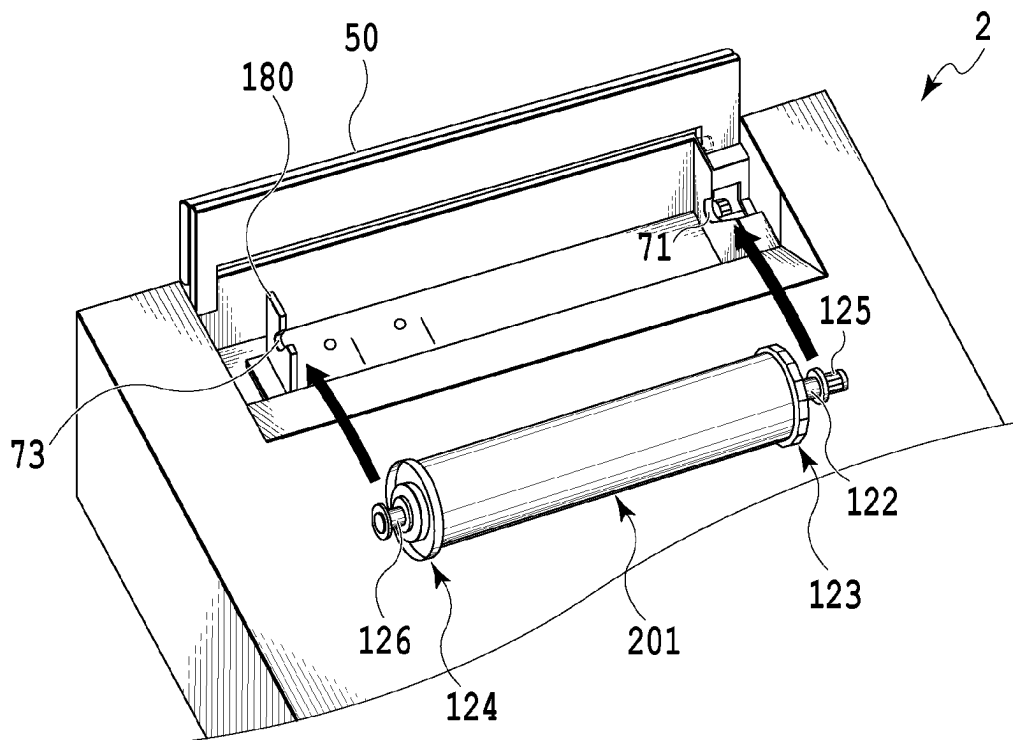


FIG. 7A

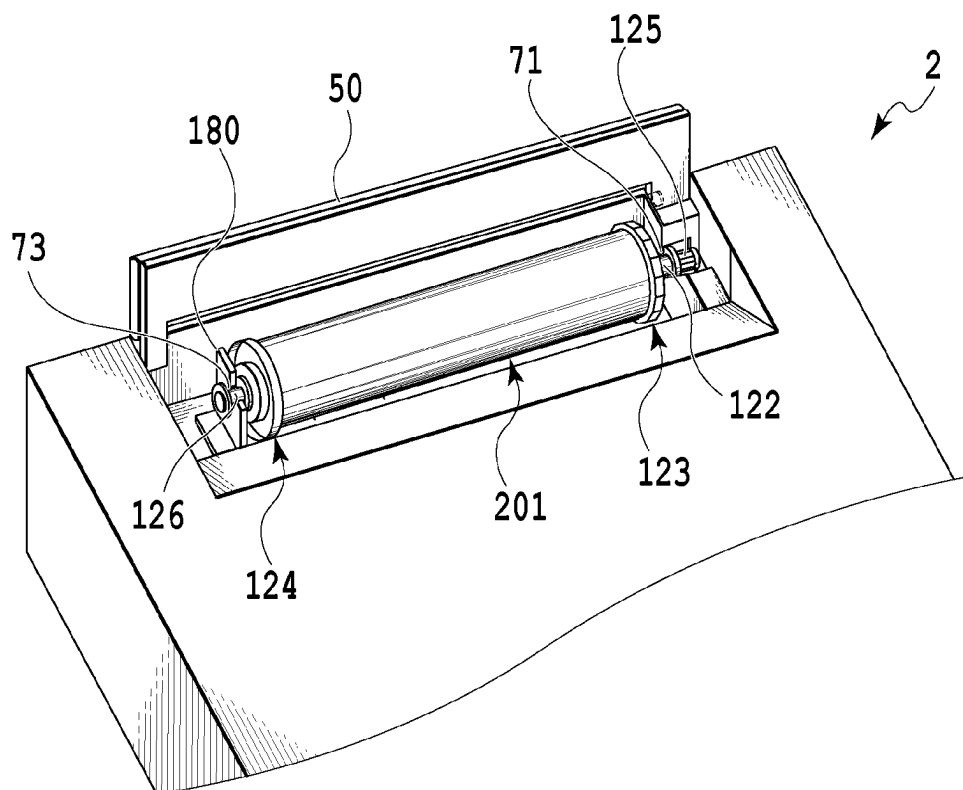


FIG. 7B

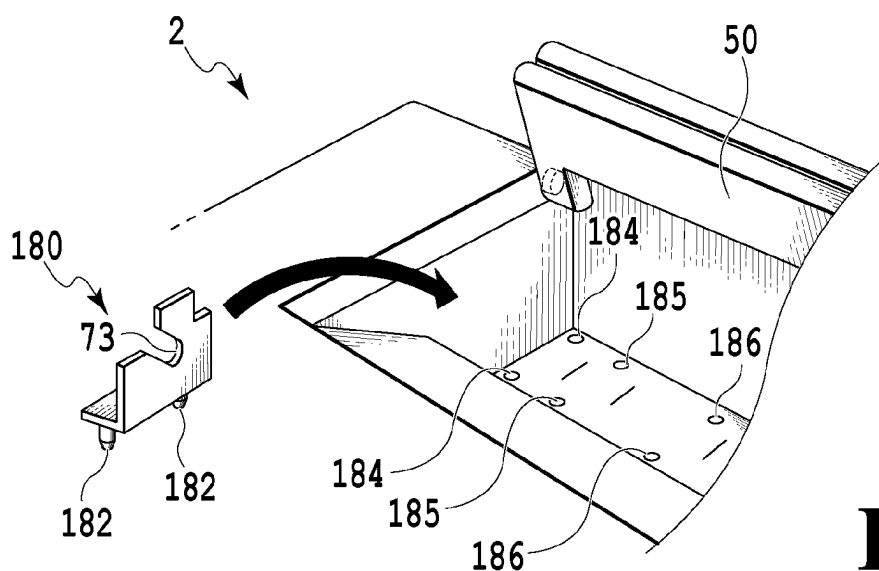


FIG. 8A

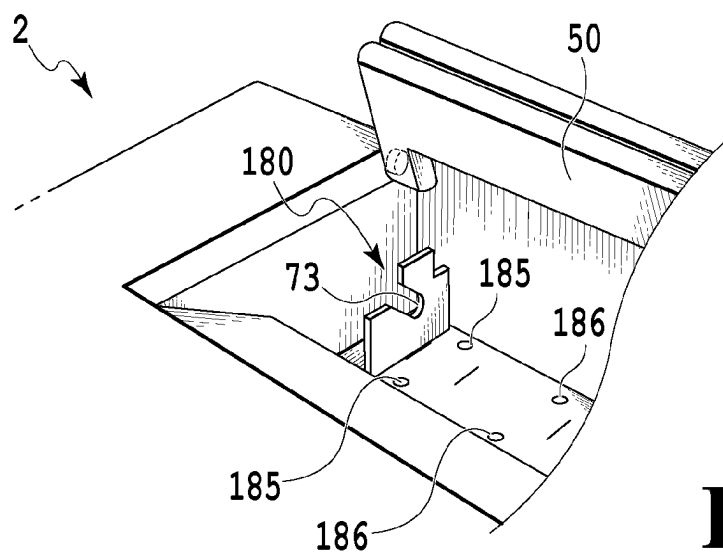


FIG. 8B

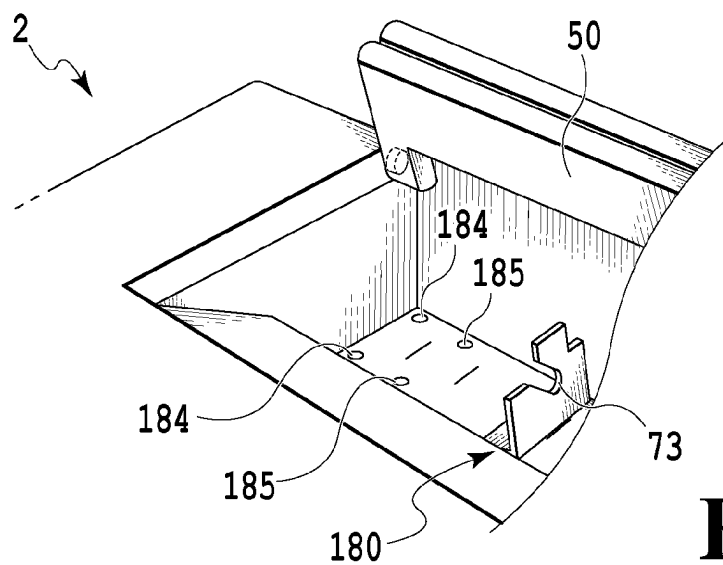


FIG. 8C

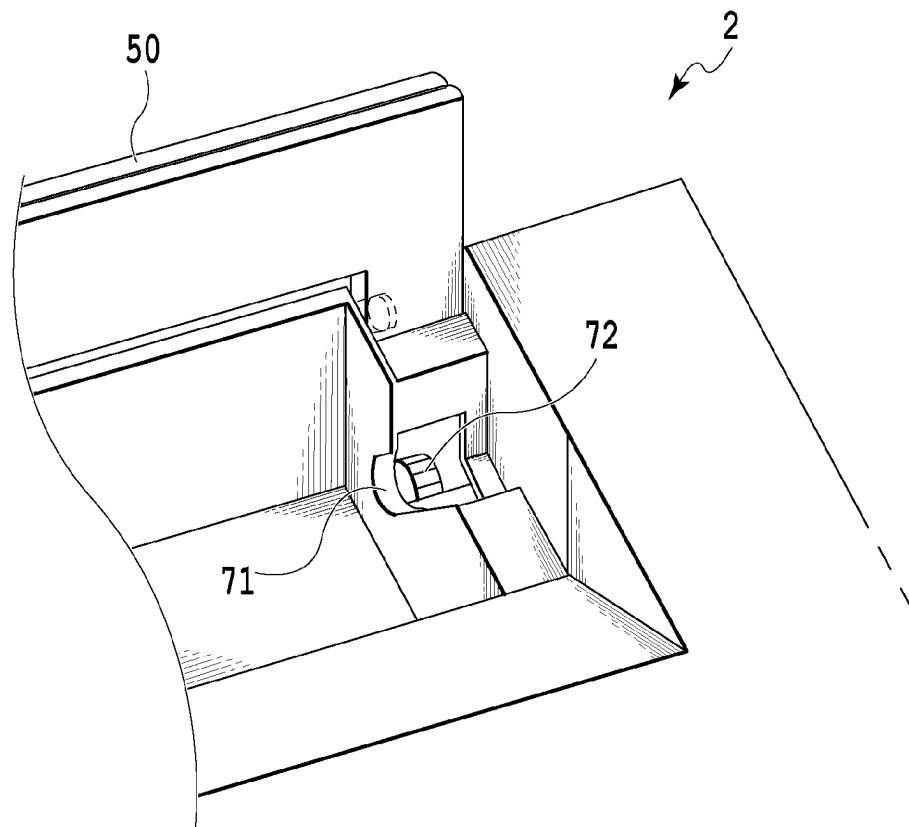


FIG.9

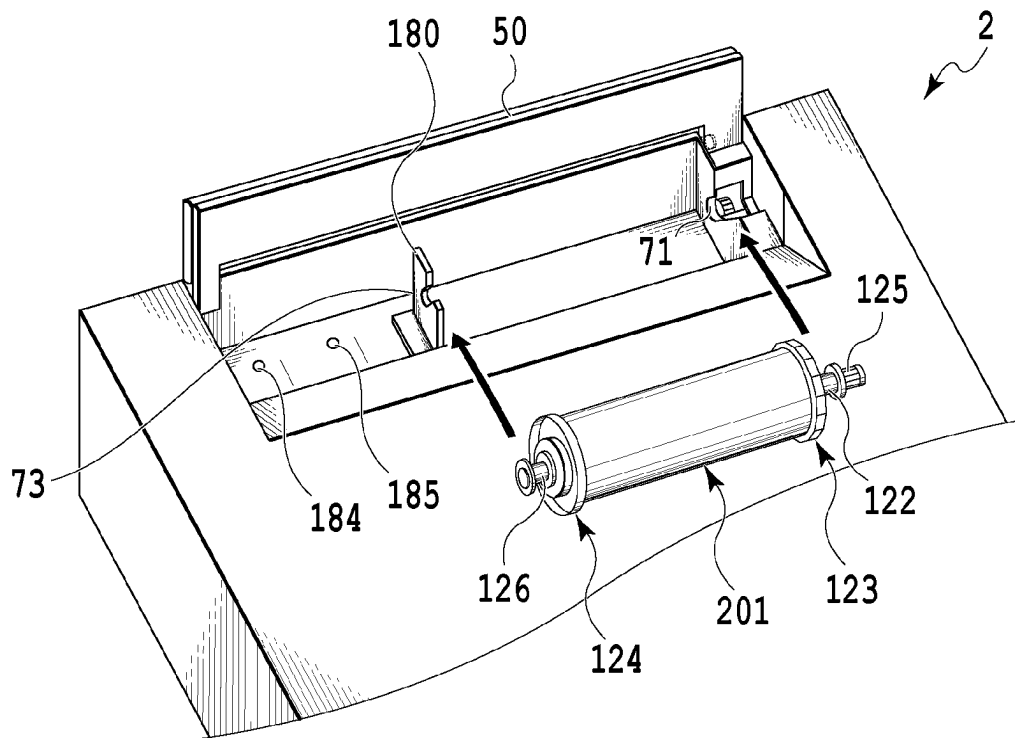


FIG.10A

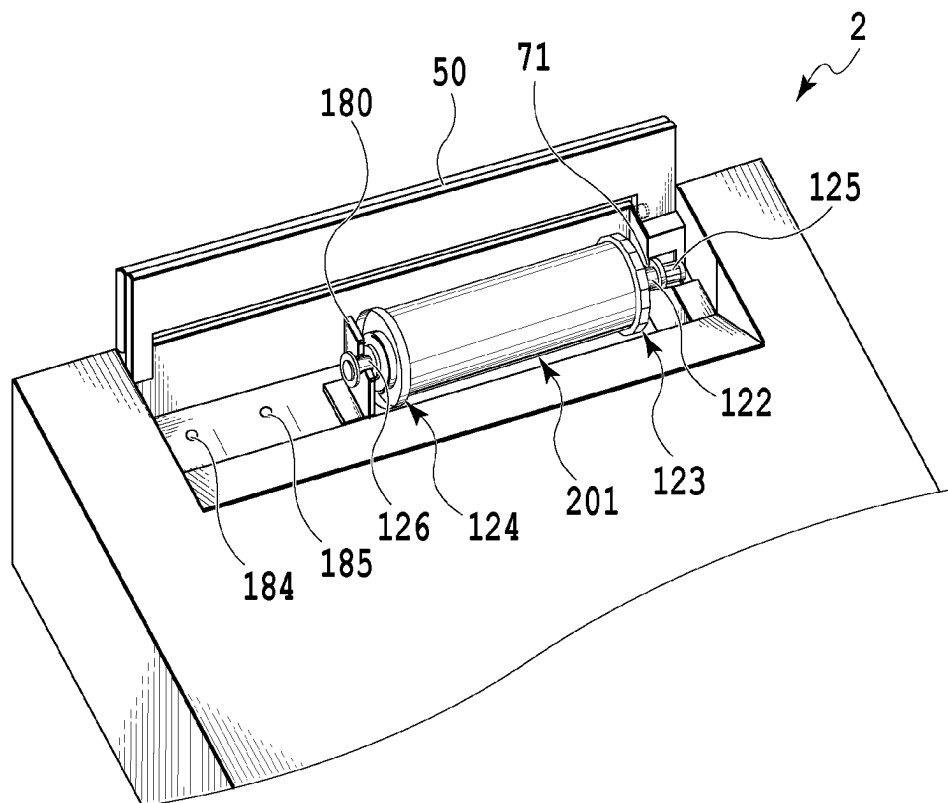


FIG.10B

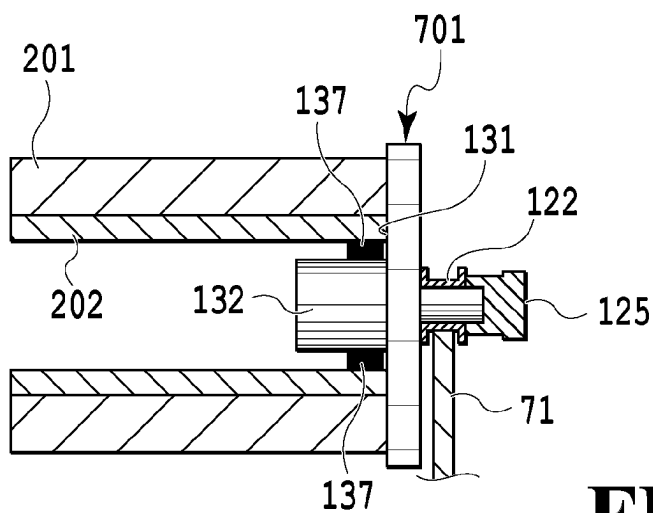


FIG.11A

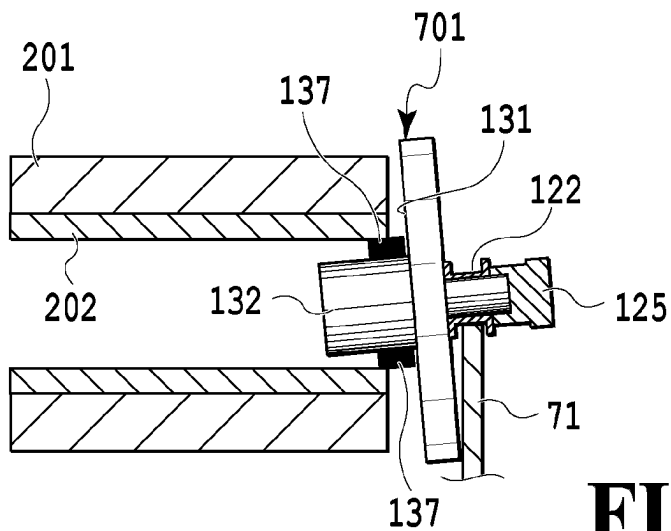


FIG.11B

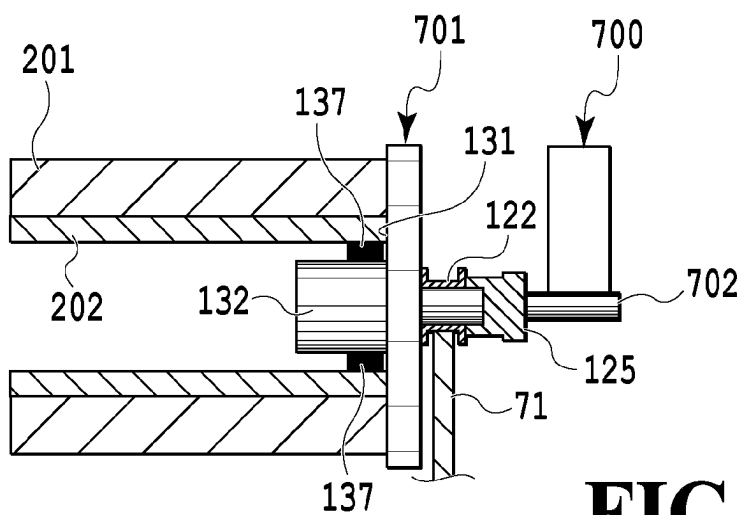
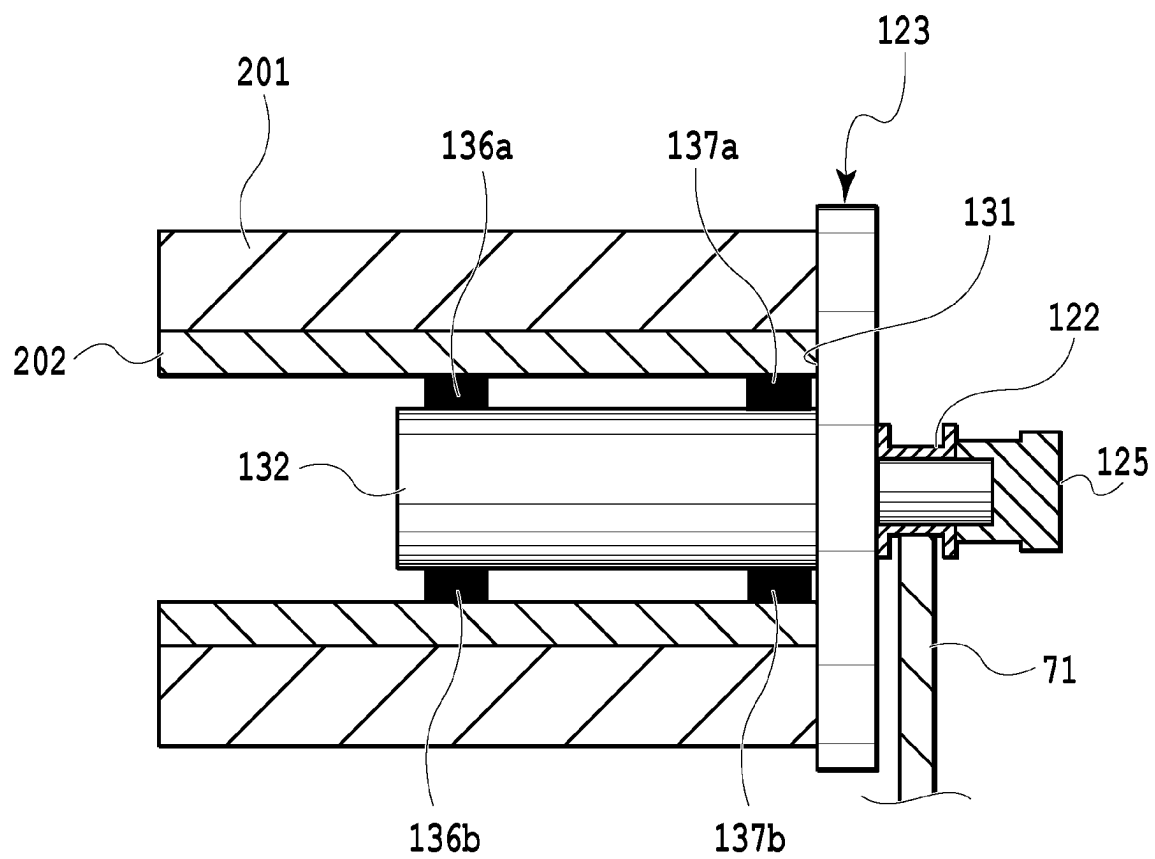


FIG.11C

**FIG.12**

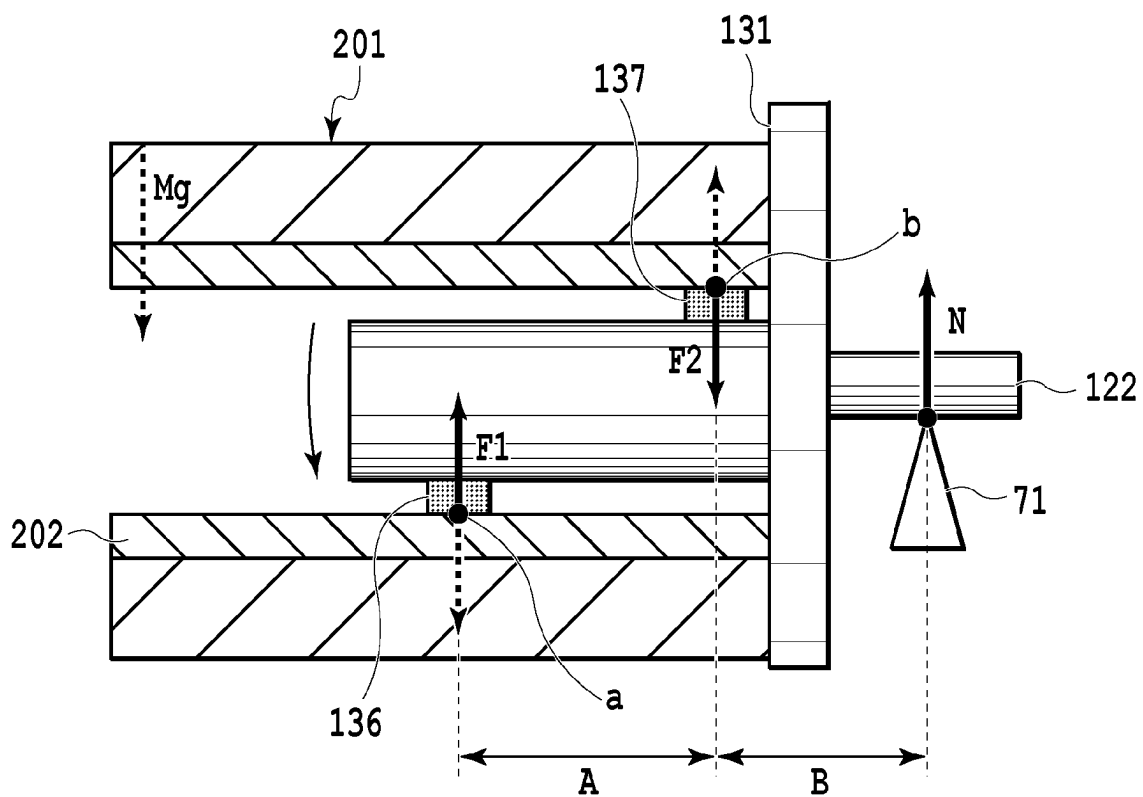


FIG.13

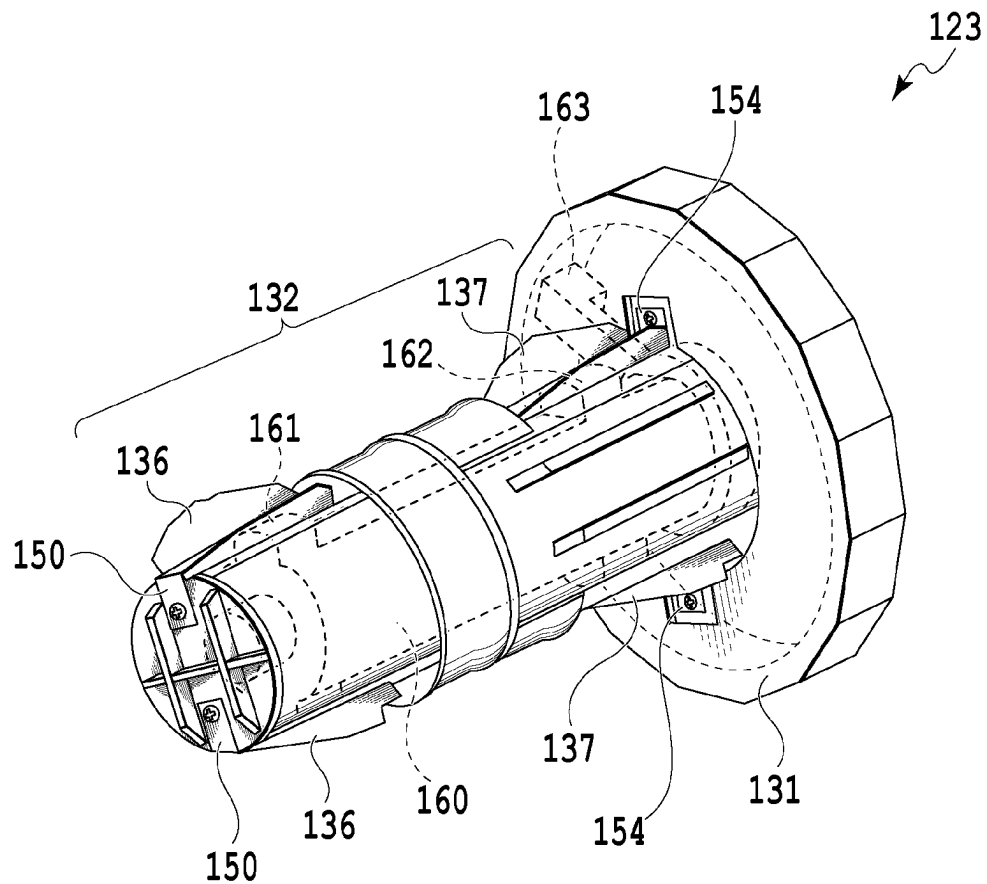
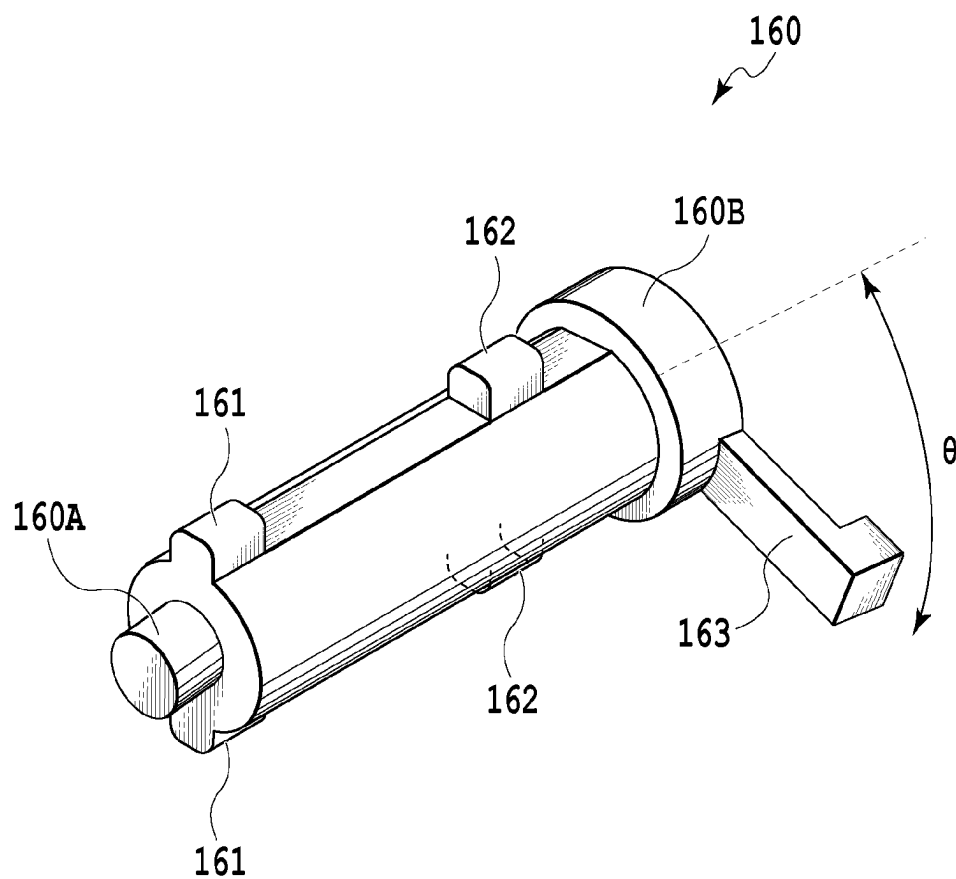


FIG.14

**FIG.15**

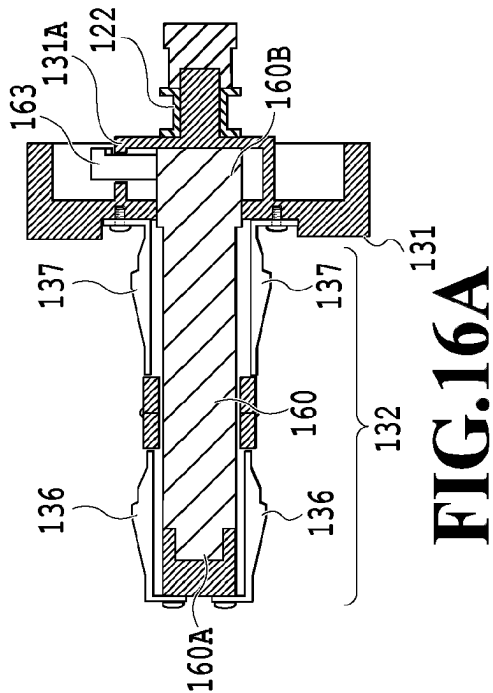


FIG. 16A

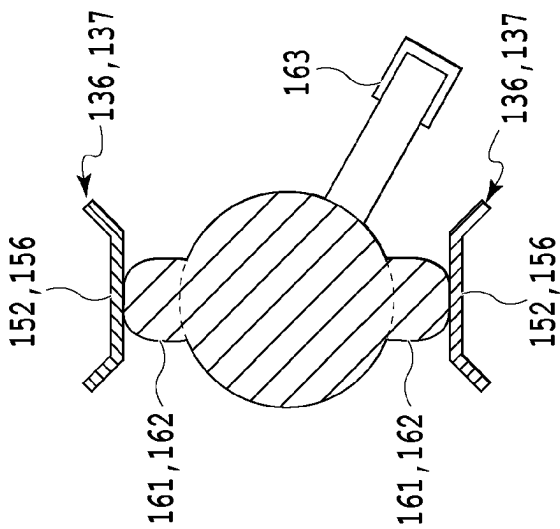
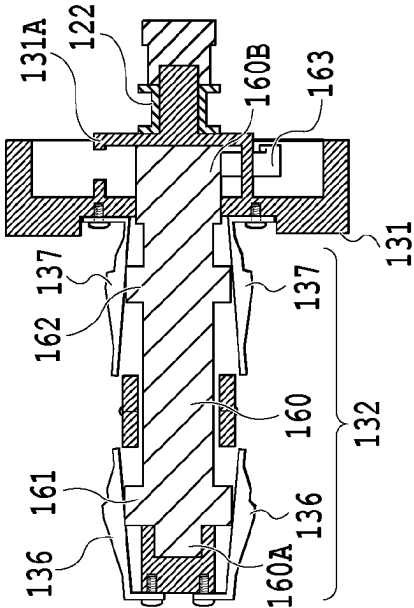


FIG. 16C

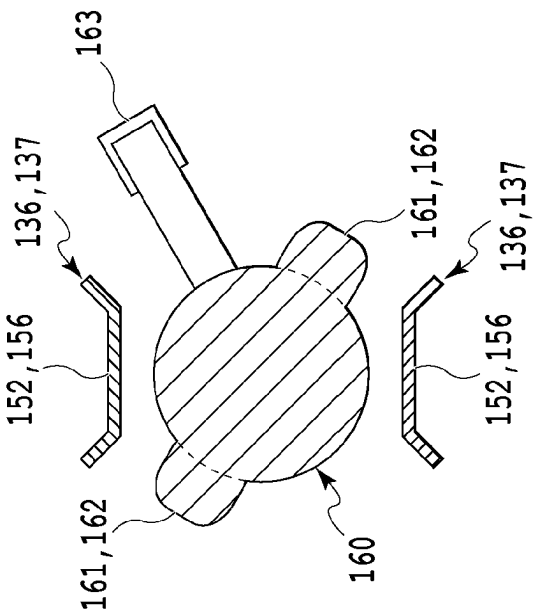


FIG. 16D

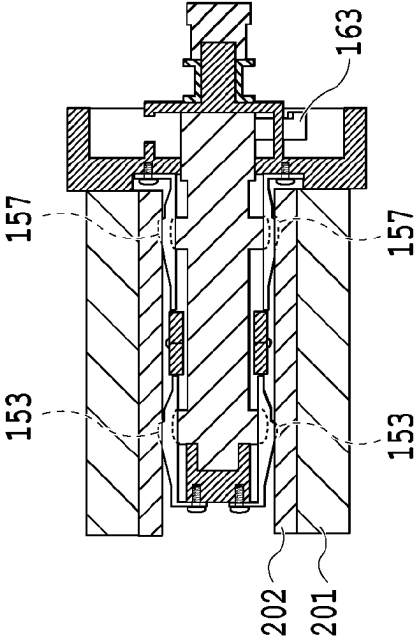


FIG. 17A

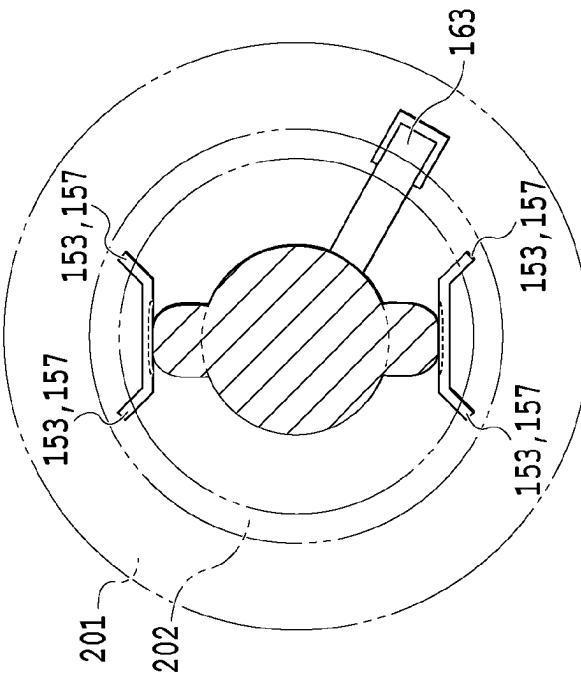


FIG. 17B

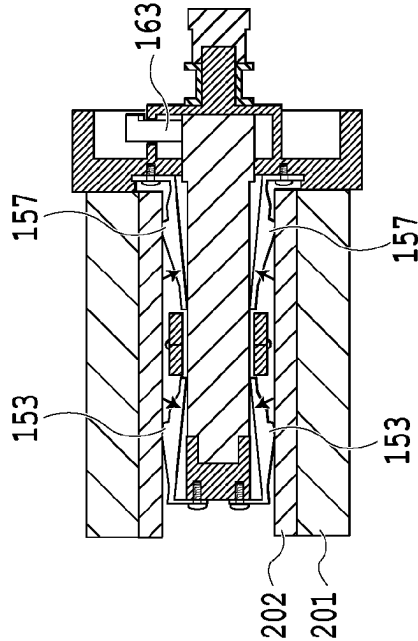


FIG. 17C

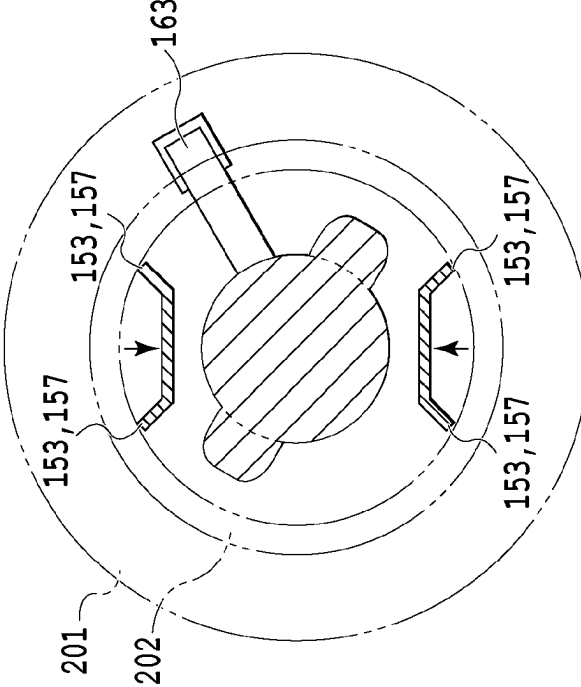


FIG. 17D

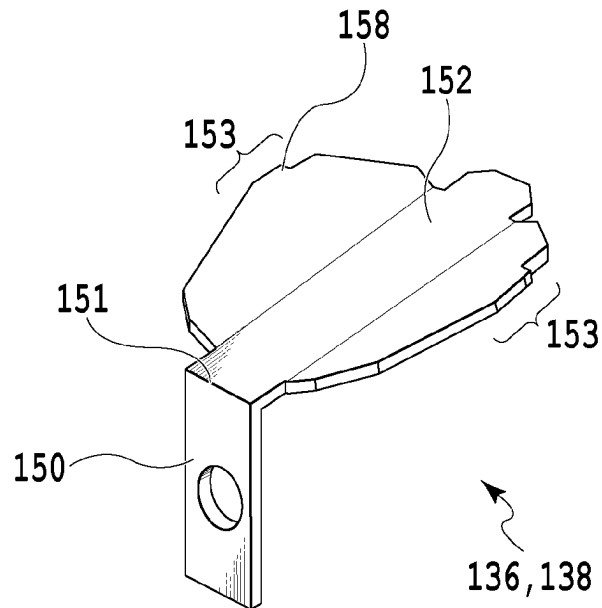


FIG. 18A

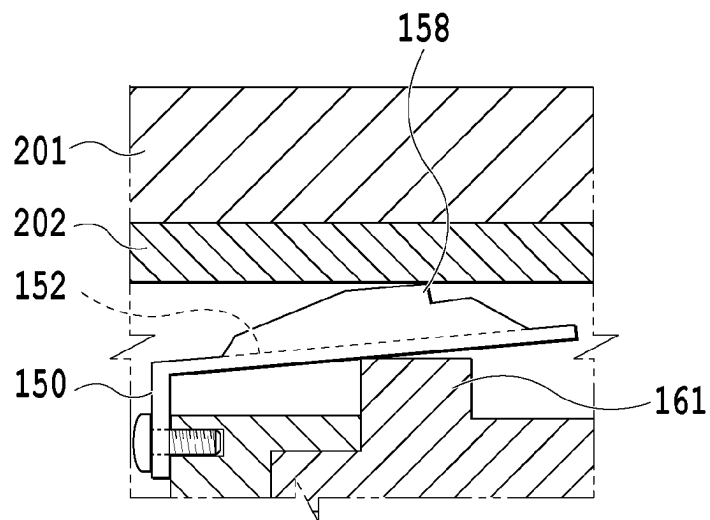
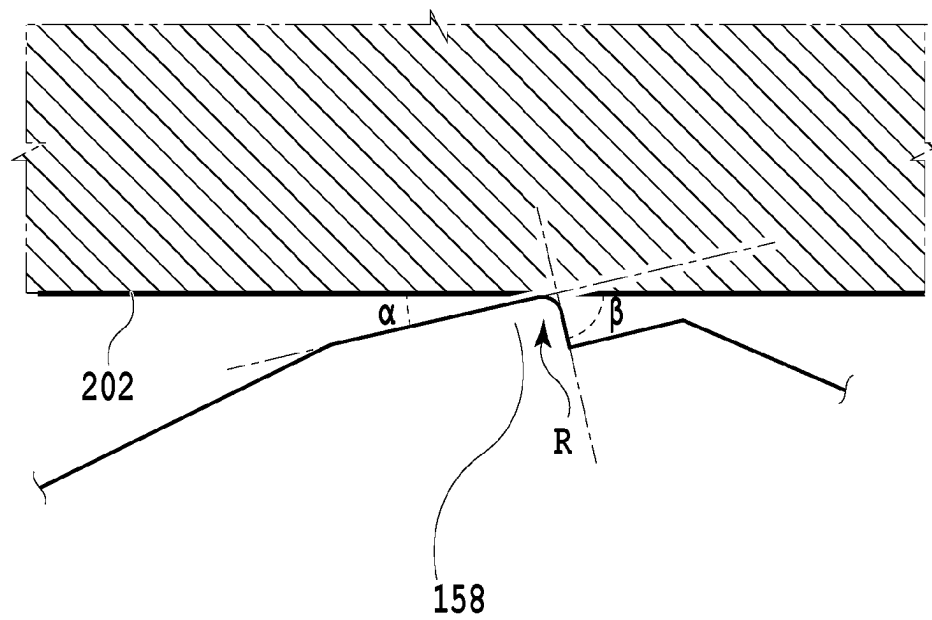


FIG. 18B

**FIG.19**

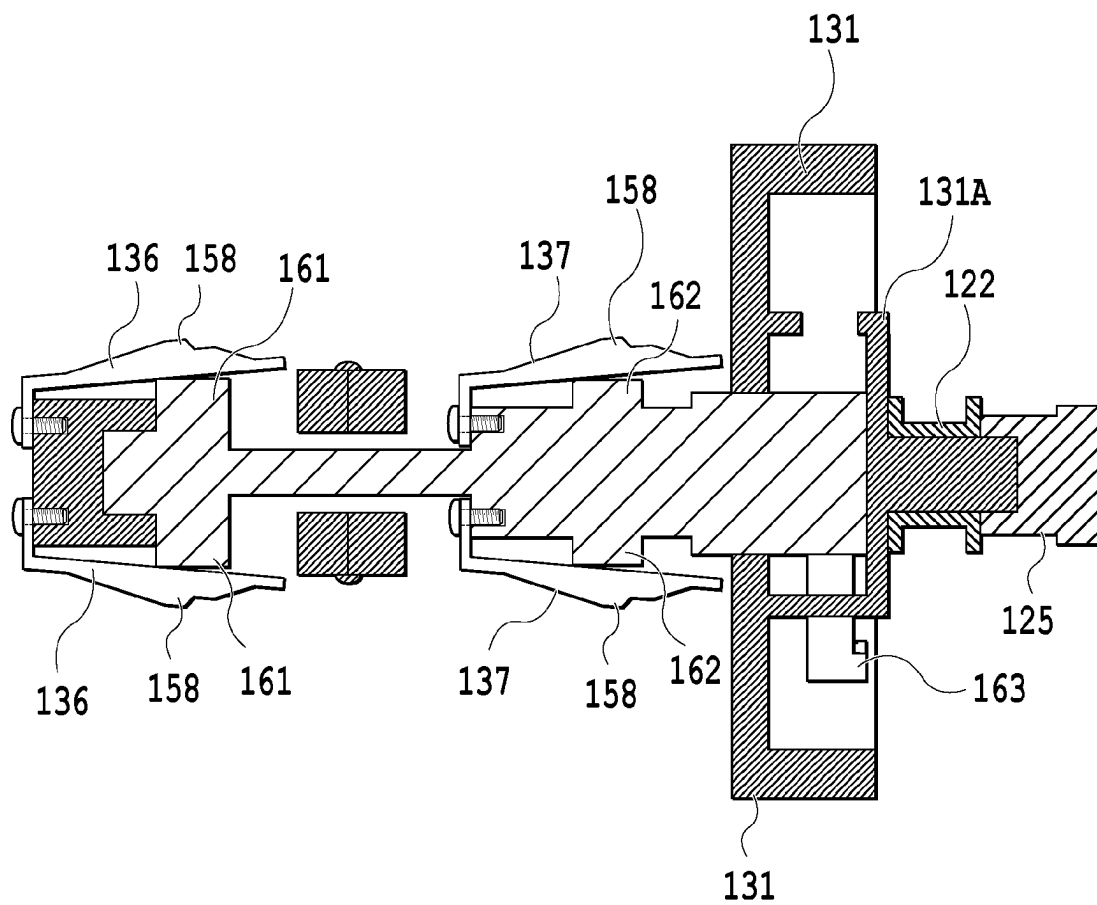


FIG.20

1

PRINTING APPARATUS AND ROLL HOLDING DEVICE

BACKGROUND OF THE INVENTION

Field of the Invention

The present disclosure relates to a printing apparatus and a roll holding device and, to be specific, relates to a structure of a roll holding device configured to hold a printing medium in the shape of a roll.

Description of the Related Art

As this type of a roll holding device, Japanese Patent No. 5553570 (hereinafter, referred to as PTL 1) describes a device that holds roll paper at two ends of a core pipe around which the roll paper is rolled up by fitting portions protruding from flanges to a hollow portion of the core pipe. According to this holding device, even in a case where the roll paper width is great, a wide working space is unnecessary for mounting of the roll paper, and it is possible to improve the workability. Additionally, since the holding device can be formed of the flanges and the fitting portions protruding therefrom, it is possible to relatively save the weight of the holding device.

In such a holding device described in PTL 1, a leaf spring is provided to each of the above-described fitting portions inserted into the roll paper core pipe, and the roll paper is held by the elastic force of this leaf spring. For this reason, the elastic force of the leaf spring may be a relatively great resistance to the insertion at the time of inserting the fitting portion of the flange into the roll paper core pipe. Particularly, the inner diameter of the core pipe may be relatively small due to the size variation of the roll paper depending on the roll paper to be used, and accordingly the insertion resistance is increased.

SUMMARY OF THE INVENTION

An aspect of the present disclosure is a printing apparatus that performs printing on a sheet drawn out from a roll in which the sheet is rolled up around a core pipe, comprising: a roll holding unit configured to hold the roll; a support unit configured to be able to rotatably support the roll holding unit; a core portion provided to the roll holding unit and configured to be inserted into a hollow portion of the core pipe from an end portion of the core pipe so as to extend along an axis of the core pipe; a contact member provided to the core portion and configured to be able to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and a cam member disposed inside the core portion and configured to be able to pivot to a first position in which the contact member is pressed onto the inner wall and to a second position in which the contact member is retracted from the inner wall.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an external perspective view schematically illustrating a printing apparatus according to one embodiment of the present disclosure;

2

FIG. 2 is a schematic cross-sectional view of the printing apparatus illustrated in FIG. 1 viewed along an X direction in FIG. 2;

FIG. 3 is a block diagram illustrating a control configuration of the printing apparatus illustrated in FIGS. 1 and 2;

FIG. 4 is a perspective view describing mounting of a roll paper unit on the printing apparatus according to one embodiment of the present disclosure;

FIGS. 5A and 5B are perspective views describing attaching of flange units to roll paper rolled up around a core pipe according to one embodiment of the present disclosure;

FIGS. 6A and 6B are perspective views illustrating fitting members according to the embodiment of the present disclosure;

FIGS. 7A and 7B are diagrams describing mounting of the roll paper unit on the printing apparatus according to one embodiment of the present disclosure;

FIGS. 8A to 8C are perspective views describing position setting of a bearing member at the time of mounting the roll paper unit according to one embodiment of the present disclosure;

FIG. 9 is a perspective view describing mounting of the roll paper unit on a conveyance-reference side according to one embodiment of the present disclosure;

FIGS. 10A and 10B are perspective views describing mounting of the roll paper unit with the smallest roll paper width according to one embodiment of the present disclosure;

FIGS. 11A to 11C are diagrams describing action of the fitting members according to a comparative example;

FIG. 12 is a diagram describing action of the fitting members according to one embodiment of the present disclosure;

FIG. 13 is a diagram describing action of the fitting members according to one embodiment of the present disclosure with parameters such as a distance between the fitting members;

FIG. 14 is a perspective view illustrating a flange unit according to another embodiment of the present disclosure;

FIG. 15 is a perspective view illustrating details of a cam member 160 illustrated in FIG. 14;

FIGS. 16A to 16D are diagrams describing an engagement relationship between the cam member and the flange unit and pressing and retraction of the fitting members by pivoting of the cam member according to the other embodiment of the present disclosure;

FIGS. 17A to 17D are diagrams describing a configuration related to elastic deformation of the fitting members according to the other embodiment of the present disclosure;

FIGS. 18A and 18B are diagrams describing the shape and action of the interference portions of the fitting member according to another additional embodiment of the present disclosure;

FIG. 19 is a diagram illustrating details of a corner portion in the interference portion illustrated in FIG. 18; and

FIG. 20 is a cross-sectional view illustrating a structure of the flange unit according to the other additional embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

Embodiments of the present disclosure are described below in details with reference to the drawings. Note that, the same reference numerals between the drawings indicate the same or corresponding elements.

Embodiment 1

FIG. 1 is an external perspective view schematically illustrating a printing apparatus according to one embodi-

3

ment of the present disclosure. A printing apparatus **2** has a form in which the elements of the printing apparatus **2** are covered with an exterior case and a cover. An operation panel **6** is provided to the exterior case on the device front surface side (a Y direction in FIG. 1), and a user can perform various settings, input commands, and confirm information on the printing apparatus **2** through this panel, for example. A roll paper cover **50** is provided at the rear of the device so as to be opened and closed by using a shaft at a device rear end as a pivot shaft. The user can mount and unmount a roll paper unit **201**, that is a roll-shaped printing medium, on and from a roll paper setting unit of the printing apparatus **2** while the cover **50** is opened. Additionally, it is possible to prevent the roll paper unit **201** from being covered with dust and trash by closing the cover **50**. As described later in details in FIG. 8, the printing apparatus **2** of the present embodiment can use roll paper portions with multiple roll paper widths, and the roll paper setting unit has corresponding mounting configurations. Two upper covers **24** and **25** that are opened and closed with a pivot shaft on the rear side are provided at the front of the device. It is possible to perform an operation to set the roll paper to a conveyance path, device maintenance for solving paper jam and the like, an attachment and detachment operation for ink tank replacement, and the like while the upper covers **24** and **25** are opened.

FIG. 2 is a schematic cross-sectional view of the printing apparatus illustrated in FIG. 1 viewed from an X direction in FIG. 2.

The printing apparatus **2** of the present embodiment can use roll paper **201S** using the roll paper unit **201** and cut paper (not illustrated) as a printing medium (a sheet). The roll paper unit **201** is formed by rolling up the roll paper **201S** around a hollow core pipe and attaching thereto flanges for holding the roll, which are described later in FIG. 5 and followings, on two sides of the core pipe. In a case where the roll paper **201S** is used as the printing medium, the roll paper unit **201** is held by the device main body side by means of a conveyance-reference side flange sliding shaft portion **122** and a non-conveyance-reference side flange sliding shaft portion **126** (see FIGS. 5A and 5B) of the unit **201**. Specifically, the roll paper unit **201** is rotatably held by a conveyance-reference side flange bearing **71** and a non-conveyance-reference side flange bearing **73** (see FIGS. 7A and 7B) of the device main body. The roll paper **201S** drawn by the user from the roll paper unit **201** held by the bearings **71** and **73** is conveyed to the downstream side through a sheet conveyance path along a lower guide **8**. Once a leading end of the roll paper **201S** reaches a nip portion between a conveyance roller **9** and a driven roller **10**, the conveyance roller **9** is driven by a conveyance motor **51** (see FIG. 3) to rotate, and accordingly the roll paper **201S** is nipped by the conveyance roller **9** and the driven roller **10**. Additionally, the roll paper **201S** is conveyed onto a platen **11** facing a printing head **13** by driving the conveyance roller **9** to rotate in the same state. On the other hand, in a case where the cut paper is used, a folded tray (not illustrated) is unfolded and arrayed on the lower guide **8** while the roll paper unit **201** is detached from the device **2**. The cut sheet is disposed on the tray in this state such that a leading end reaches the nip portion between the conveyance roller **9** and the driven roller **10**.

The platen **11** forms an additional conveyance path for the conveyed roll paper **201S** (or the cut paper; the same applies to the description of FIG. 2 below). Also, the platen **11** supports the roll paper from the back side and maintains a constant distance between the printing head **13** and the roll

4

paper **201S**. The printing head **13** is built in a carriage **12**, and the carriage **12** is guided and supported by a carriage shaft **14** as a scanning guide extending in the X direction to move reciprocally in $\pm X$ directions (main scanning directions) along the scanning guide. The printing head **13** includes ejection port arrays in which multiple ejection ports (nozzles) ejecting inks of yellow (Y), cyan (C), magenta (M), and black (Bk) are arrayed in the Y direction, for example. The inks are ejected from those ejection ports of the ejection port arrays in a downward direction (a-Z direction) in accordance with image data while the carriage **12** is moving. Once an image of one line is printed by the ejection operation of the printing head **13** and the movement of the carriage **12** as described above, the roll paper **201S** is conveyed by a predetermined amount in the conveyance direction Y by the conveyance roller **9** and the driven roller **10**, and the carriage **12** is moved again to print the next line. An image is printed on the entirety of a predetermined page by repeating those operations. Ink tanks respectively corresponding to the inks Y, M, C, and Bk ejected from the printing head **13** are provided to a right end (an end portion in the -X direction in FIG. 1) of the device **2** in the movement direction of the carriage **12**. The printing head **13** is supplied with the inks from those ink tanks through corresponding tubes. A cutter **16** is provided to the roll paper conveyance path at a predetermined distance on the downstream side of the printing head **13** and can cut the roll paper **5201** into a predetermined length. Note that, the printing operation in a case of using the cut paper as the printing medium is similar to that in the case of using the roll paper **201S**. The cut roll paper **201S** after printing ends is conveyed to a paper discharging tray **18** (see FIG. 1).

FIG. 3 is a block diagram illustrating a control configuration of the printing apparatus illustrated in FIGS. 1 and 2.

The printing apparatus **2** of the present embodiment includes a control unit **402**, and the control unit **402** controls the printing operation and the like based on a printing command transmitted from a host device **401**. Specifically, the host device **401** transmits the printing command of an image generated by the host device **401** to a main control unit **403** of the printing apparatus **2**. The control unit **402** includes the main control unit **403** and an image printing control unit **404** as a main configuration. The main control unit **403** includes a CPU **406**, a ROM **407**, and a RAM **408**, and the CPU **406** controls overall the printing apparatus **2** according to various programs and parameters stored in the ROM **407** by using the RAM **408** as a working area. The image printing control unit **404** controls driving of the conveyance motor **51**, a carriage motor **54**, the printing head **13**, an actuator of the cutter **16**, a roll paper motor **55**, and the like under instruction of the main control unit **403**.

Note that, in addition to the above-described form, the form of the printing apparatus includes a printing unit in a copier or a facsimile, for example. Additionally, a multi-function peripheral to which other functions such as a printer function, a facsimile function, an original document reading function, and an imaging function are added and also a system device formed by connecting the printing apparatus to a host device such as a computer are included in the printing apparatus. It is obvious from the following descriptions that it is possible to apply the present disclosure to those devices, and in this regard, those devices are included in the printing apparatus.

[Mounting of Roll Paper Unit]

FIG. 4 is a perspective view describing mounting of the roll paper unit **201** on the printing apparatus **2** according to one embodiment of the present disclosure. At the time of

5

mounting the roll paper unit **201**, flange units (described later) are attached to two ends of a core pipe **202** around which the roll paper **201S** is rolled up. This core pipe **202** may be made of paper such as cardboard. The roll paper unit **201** to which the flange units are attached as described above can be inserted and mounted on a roll paper setting unit while the roll paper cover **50** of the printing apparatus **2** is opened by the user. As illustrated in FIGS. 7A and 7B, the sliding shaft portions **122** and **126** of the flange units that function as a roll holding member (a roll holding device) in this process are supported by the conveyance-reference side flange bearing **71** and the non-conveyance-reference side flange bearing **73** of the roll paper setting unit. Here, the conveyance-reference side flange bearing **71** and the non-conveyance-reference side flange bearing **73** function as a support unit.

[Attaching of Flange Units to Roll Paper Portion]

FIGS. 5A and 5B are perspective views describing attaching of the flange units into the roll paper **201S** rolled up around the core pipe **202**.

The flange units function as a roll holding unit (a roll holding device) and include a conveyance-reference side flange unit **123** and a non-conveyance-reference side flange unit **124**. The flange units **123** and **124** are respectively inserted and attached to the corresponding two ends of the hollow portion of the roll paper core pipe **202**. To be specific, as illustrated in FIG. 5A, the conveyance-reference side flange unit **123** includes a conveyance-reference side flange **131** and a conveyance-reference side flange core portion **132** protruding from one surface of the conveyance-reference side flange **131**. Additionally, the conveyance-reference side flange unit **123** includes the conveyance-reference side flange sliding shaft portion **122** protruding from a surface of the conveyance-reference side flange **131** on the opposite side of the above-described one surface and a flange gear **125** continuous from the conveyance-reference side flange sliding shaft portion **122**. Here, the surface of the conveyance-reference side flange **131** on the conveyance-reference side flange core portion **132** side is a flat surface, and with this surface, the conveyance-reference side flange **131** can be abutted (particularly, closely abutted) onto an end portion of the roll paper core pipe **202** and an end surface of the roll paper **201S** rolled up around the roll paper core pipe **202**. The conveyance-reference side flange core portion **132** includes first fitting members **136** and second fitting members **137** disposed at a predetermined distance from each other in a longitudinal direction thereof. As described later in details in FIG. 12 and the like, those first fitting member **136** and second fitting member **137** can suppress tilting of the flange unit attached to the roll paper unit **201** due to the weight of the roll paper **201S**.

As with the conveyance-reference side flange unit **123**, the non-conveyance-reference side flange unit **124** includes a non-conveyance-reference side flange **135** and a non-conveyance-reference side flange core portion **134** protruding from one surface of the non-conveyance-reference side flange **135**. Additionally, the non-conveyance-reference side flange unit **124** includes a non-conveyance-reference side flange sliding shaft portion **126** protruding from the other surface of the non-conveyance-reference side flange **135**. Here, the surface of the non-conveyance-reference side flange **135** on the non-conveyance-reference side flange core portion **134** side is a convexly curved surface. For this reason, the surface can be abutted onto the roll paper core pipe **202** and the end surface of the roll paper **201S** rolled up around the roll paper core pipe **202** and can bias the roll paper unit **201** to the conveyance-reference side. As with the

6

conveyance-reference side, the non-conveyance-reference side flange core portion **134** includes first fitting members **138** and second fitting members **139** disposed at a predetermined distance from each other. As described later in details, those first fitting members **138** and second fitting members **139** suppress tilting of the flange unit due to the weight of the roll paper **201S**.

In the above-described configuration, the pair of the fitting member **136** on the conveyance-reference side are disposed at intervals of 180 degrees along a circumference of the flange core portion **132**. The pair of the fitting member **137** on the conveyance-reference side are also disposed at intervals of 180 degrees along a circumference of the flange core portion **132**. Similarly, the pair of the fitting member **138** on the non-conveyance-reference side are disposed at intervals of 180 degrees along a circumference of the flange core portion **134**. The pair of the fitting member **139** on the non-conveyance-reference side are also disposed at intervals of 180 degrees along a circumference of the flange core portion **134**. One of the pair of fitting members **136** and one of the pair of the fitting member **137** are disposed at the same positions along circumferences of the flange core portion **132**. This also applies to the other of the pair of fitting member **136** and the other of the pair of the fitting member **137**. Similarly, one of the pair of fitting members **138** and one of the pair of the fitting member **139** are disposed at the same positions along circumferences of the flange core portion **134**. This also applies to the other of the pair of fitting member **138** and the other of the pair of the fitting member **139**. As a result, each of flange core portions **132**, **134** includes the four fitting members. Note that, the number of the fitting members is not limited thereto and can be determined in accordance with specifications of the device and the like. Likewise, it is not limited to the mode that the number of the fitting members **137**, **139** respectively close to the flange **131**, **135** and the number of the fitting members **136**, **138** respectively far from the flange **131**, **135** in their respective flange core portion are equal to each other.

FIGS. 6A and 6B are perspective views illustrating the fitting members according to the embodiment of the present disclosure. FIG. 6A illustrates the first fitting member **136**, **138** and FIG. 6B illustrates the second fitting member **137**, **139**. The fitting member **136**, **138** includes a flat surface portion **152** and a pair of interference portions **153** which form a pair of wings. A metallic plate member is bended along a boundary between the flat surface portion **152** and one of the pair of interference portion **153** and along another boundary between the flat surface portion **152** and the other of the pair of interference portion **153**. Similarly, the fitting member **137**, **139** includes a flat surface portion **156** and a pair of interference portions **157** which form a pair of wings. A metallic plate member is bended along a boundary between the flat surface portion **156** and one of the pair of interference portion **157** and along another boundary between the flat surface portion **156** and the other of the pair of interference portion **157**. With this bending, the fitting member **136**, **138** has the elasticity against the opposite forces acting on the flat surface portion **152** and the pair of interference portions **153**. This also applies to the fitting member **137**, **139**. The fitting members **136**, **138** are respectively bent from the flat surface portions **152** at constricted portions **151**, and the bent portion functions as a fixing portion **150** screwed on the flange unit. The fitting members **137**, **139** are respectively bent from the flat surface portions **156** at constricted portions **155**, and the bent portion functions as a fixing portion **154** screwed on the flange unit.

As illustrated in details in FIG. 14, the fixing portion 154 of the fitting member 137 is screwed on the flange 131. Likewise, as illustrated in detail in FIG. 14, the above-described fixing portion 150 of the fitting member 136 is screwed on an end surface of the flange core portion 132. Similarly, although no figure shows, the fixing portion 154 of the fitting member 139 is screwed on the flange 135. Likewise, the above-described fixing portion 150 of the fitting member 138 is screwed on an end surface of the flange core portion 134. In the present embodiment, with respect to each of fitting members 136, 137, 138, 139, the fixing portion 150 or 154 is fixed by a screw as described above, and the flat surface portion 152 or 156 is in contact with an outer surface of the corresponding flange core portion 132. Note that, an opening may be provided in a region of the flange core portion 132 corresponding to portions other than the fixing portion 150 or 154 of the fitting member, and the fitting member is deformed so as to enter the inside of a hollow portion of the flange core portion through this opening. In this case, it is favorable that the elastic force of the fitting member is relatively great enough so that the core pipe 202 of the roll paper unit 201 can be appropriately pressed and held by the elastic force.

In order to attach the conveyance-reference side flange unit 123 and the non-conveyance-reference side flange unit 124 to the roll paper portion, first, each of the conveyance-reference side flange core portion 132 and the non-conveyance-reference side flange core portion 134 is inserted into the hollow portion of the core pipe 202 of the roll paper portion. The outer diameters of both of the conveyance-reference side flange core portion 132 and the non-conveyance-reference side flange core portion 134 except for the portions of the fitting members 136 and 138 are designed smaller than the inner diameter of the hollow portion of the core pipe 202 of the roll paper portion. On the other hand, each of the first fitting members 136, the second fitting members 137, the first fitting members 138, and the second fitting members 139 has the shape of a leaf spring that will be elastically deformed as described above in FIG. 6. Thus, those fitting members 136-139 can have a dimension relationship for being fitted with the hollow portion of the roll paper core pipe 202, in other words, a dimension relationship for being in contact with the roll paper core pipe 202 with force acting on each other.

FIG. 5B illustrates a state where the conveyance-reference side flange unit 123 and the non-conveyance-reference side flange unit 124 are attached to the roll paper portion including the core pipe 202 and the roll paper rolled up around the core pipe 202. Hereinafter, a unit in this attached state is also referred to as a "roll unit".

[Mounting of Roll Paper Unit on Device Main Body]

FIGS. 7A and 7B are diagrams describing mounting of the roll paper unit on the printing apparatus 2. FIG. 7A illustrates a state before the roll paper unit, in other words, the roll paper portion to which the conveyance-reference side flange unit 123 and the non-conveyance-reference side flange unit 124 are attached, is mounted on the printing apparatus 2, and FIG. 7B illustrates a state where the roll paper unit is mounted on the printing apparatus 2.

As illustrated in FIG. 7A, the conveyance-reference side flange sliding shaft portion 122 is set to the conveyance-reference side flange bearing 71, and the non-conveyance-reference side flange sliding shaft portion 126 is set to the non-conveyance-reference side flange bearing 73. In a case where the user performs this mounting, it is possible to make positioning so as to fit the conveyance-reference side flange sliding shaft portion 122 to the conveyance-reference side

flange bearing 71 and the non-conveyance-reference side flange sliding shaft portion 126 to the non-conveyance-reference side flange bearing 73, respectively. Particularly, even in a case where the conveyance-reference side flange sliding shaft portion 122 is close to the conveyance-reference side flange bearing 71 and the non-conveyance-reference side flange sliding shaft portion 126 is close to the non-conveyance-reference side flange bearing 73, respectively, the operation can be performed while the user can see the respective fitting positions. Thus, it is possible to facilitate positioning at the time of mounting the roll paper unit 201 on the printing apparatus 2. As also illustrated in FIG. 7B, even in a state after the mounting of the roll paper unit 201 on the printing apparatus 2 is completed, the conveyance-reference side flange sliding shaft portion 122 and the conveyance-reference side flange bearing 71 can be investigated visually. Also, the non-conveyance-reference side flange sliding shaft portion 126 and the non-conveyance-reference side flange bearing 73 can be investigated visually. Accordingly, it is obvious that those constituents can be investigated visually even in the middle of mounting.

[Position Setting of Bearing Member]

FIGS. 8A, 8B, and 8C are perspective views from obliquely above describing position setting of the bearing member in a case of mounting the roll paper unit and illustrate the roll paper setting unit of the printing apparatus 2 with the opened roll paper cover 50.

FIG. 8A illustrates a state where no bearing portion is set to a non-conveyance-reference side disposing portion of the roll paper unit 201 in the printing apparatus 2. FIG. 8B illustrates a state where a non-conveyance-reference side bearing member 180 for mounting the roll paper unit 201 with a wide roll paper width on the non-conveyance-reference side disposing portion is set (attached). FIG. 8C illustrates a state where the non-conveyance-reference side bearing member 180 for mounting the roll paper unit 201 with a narrow roll paper width on the non-conveyance-reference side disposing portion is set.

In FIG. 8A, at the time of mounting the roll paper unit 201 on the printing apparatus 2, a positioning projection 182 of the non-conveyance-reference side bearing member 180 is inserted into any one of positioning holes 184, 185, and 186 corresponding to the roll paper widths in the non-conveyance-reference side disposing portion. The positioning holes 184, 185, and 186 are set in accordance with the three sizes of the roll paper widths assumed to be used in the printing apparatus 2. Note that, the positioning holes provided at three portions are indicated in the example illustrated in FIG. 8A; however, as a matter of course, positioning holes may be provided at any number of positions corresponding to the number of sizes of the roll paper width assumed to be used. In the present embodiment, the positioning hole 184 corresponding to the maximum roll paper width of the roll paper unit 201, the positioning hole 185 corresponding to a smaller roll paper width than the above, and the positioning hole 186 corresponding to a further smaller roll paper width than the above are provided. FIG. 8B illustrates a state where the non-conveyance-reference side bearing member 180 is attached to the positioning hole 184 corresponding to the greatest roll paper width. FIG. 8C illustrates a state where the non-conveyance-reference side bearing member 180 is attached to the positioning hole 186 corresponding to the smallest roll paper width.

FIG. 9 is a perspective view from obliquely above describing mounting of the roll paper unit on the conveyance-reference side and illustrates the roll paper setting unit

on the conveyance-reference side of the printing apparatus 2 with the opened roll paper cover 50.

At the time of attaching the conveyance-reference side flange unit 123 to the printing apparatus 2, the conveyance-reference side flange sliding shaft portion 122 (FIG. 7) is engaged to the conveyance-reference side flange bearing 71. In this process, the flange gear 125 (FIG. 7) is engaged with a driving gear 72 (FIG. 9) provided on the main body side of the printing apparatus 2. This driving gear 72 is rotated and driven by the roll paper motor 55 (FIG. 3) provided on the main body side of the printing apparatus 2. Thus, the conveyance-reference side flange unit 123 can rotate in forward direction and reverse direction with the roll paper unit 201 and the non-conveyance-reference side flange unit 124, whereby a feeding operation by the roll paper unit 201 is performed.

FIGS. 10A and 10B are perspective views describing mounting of the roll paper unit with the smallest roll paper width. FIG. 10A illustrates a state before the roll paper unit 201 is mounted on the printing apparatus 2, and FIG. 10B illustrates a state after the roll paper unit 201 is mounted on the printing apparatus 2. As illustrated in FIG. 10A, as with the mounting in a case where the roll paper unit 201 has a great roll paper width, the conveyance-reference side flange sliding shaft portion 122 is engaged to the conveyance-reference side flange bearing 71, and the non-conveyance-reference side flange sliding shaft portion 126 is engaged to the non-conveyance-reference side flange bearing 73. Accordingly, ease of mounting the roll paper unit 201 on the printing apparatus is similar to the case where the roll paper width is wide. As illustrated in FIG. 10B, even after the roll paper unit 201 is mounted on the printing apparatus 2, the conveyance-reference side flange sliding shaft portion 122 and the conveyance-reference side flange bearing 71 can be investigated visually. Also, the non-conveyance-reference side flange sliding shaft portion 126 and the non-conveyance-reference side flange bearing 73 can be investigated visually similarly to the case of mounting of the roll paper unit 201 with a wide paper width, and mounting of the roll paper unit 201 on the printing apparatus is facilitated.

[Action of Fitting Members in Flange Unit]

The fitting members according to one embodiment of the present disclosure suppress each flange unit from tilting due to the weight of the roll paper portion. FIGS. 11A to 11C are diagrams describing action of the fitting members according to a comparative example, and FIG. 12 is a diagram describing action of the fitting members according to one embodiment of the present disclosure. Any of FIGS. 11A to 11C and 12 illustrate a state where the roll paper unit 201 formed by attaching the flange unit to the roll paper portion is mounted on the printing apparatus 2.

As illustrated in FIG. 11A, a flange unit 701 according to the comparative example includes the pair of the fitting members 137 on the flange core portion 132. The conveyance-reference side flange sliding shaft portion 122 of the flange unit 701 is slidably supported by the conveyance-reference side flange bearing 71 on the device main body side. In this case, the entirety of the flange unit 701 may tilt as illustrated in FIG. 11B depending on the weight of the roll paper unit 201. If the roll paper is fed from the roll paper unit 201 in such a state, inconvenience occurs such as displacement of an end portion position of the fed roll paper. To deal with this, as illustrated in FIG. 11C, there may be considered a configuration in which a shaft portion 702 extending from the flange unit 701 is provided, and also a member 700 that applies acting force in the opposite direction of the tilting direction to the shaft portion 702 is provided on the printing

apparatus 2 side. However, in a case of this configuration, there are needed a mechanism that retracts the member 700 from the extending shaft portion 702 of the flange unit at the time of attaching the flange unit, a mechanism that moves the member 700 to a pressing position after the flange unit is attached, and the like. For this reason, there is inconvenience that the configuration of the main body device side is complicated, and the cost is increased.

To deal with this, in one embodiment the present disclosure, fitting members 137a and 137b (second contact members) and fitting members 136a and 136b (first contact members) apart from each other by a predetermined distance are provided to the conveyance-reference side flange core portion 132 as illustrated in FIG. 12. Thus, each of the fitting members 137a and 137b and the fitting members 136a and 136b is put in contact with the core pipe 202 of the roll paper unit 201 with no clearance. Additionally, the distance between the fitting members 137a, 137b and the fitting members 136a, 136b causes the moment against the moment causing the flange unit to tilt, and thus the tilting can be suppressed.

Note that, the fitting members 136 and the fitting members 137 may be provided to be distributed over the entire circumference of the inner wall surrounding the hollow portion of the core pipe 202. However, in this configuration, the cost is increased, and the force required to insert the conveyance-reference side flange unit 123 into the core pipe 202 of the roll paper portion is increased. For this reason, in the present embodiment, the fitting portions are not provided over the entire circumference, but instead the fitting members 136a and 136b are provided at two portions at an interval of 180 degrees in the circumferential direction. Likewise, the fitting members 137a and 137b are provided at two portions at an interval of 180 degrees in the circumferential direction. Here, each of the fitting members 136a and 136b has the two interference portions 153 in the shape of wings, and each of the fitting members 137a and 137b has the two interference portions 157 in the shape of wings. Accordingly, the four interference portions 153 and the four interference portions 157 in total are put in contact with the inner wall surrounding the hollow portion of the core pipe 202. Accordingly, regarding the conveyance-reference side flange 123, the total number of the interference portions (the interference portions 153 and the interference portions 157) put in contact with the inner wall surrounding the hollow portion of the core pipe 202 is eight.

As above, the situation described above with reference to FIGS. 11A to 11C and 12 is similarly applied to the flange unit on the non-conveyance-reference side. That is, regarding the non-conveyance-reference side flange 124, each of the fitting members 138a and 138b includes the two interference portions 153 in the shape of wings, and each of the fitting members 139a and 139b includes the two interference portions 157 in the shape of wings. Accordingly, regarding the non-conveyance-reference side flange 124, the total number of the interference portions (the interference portions 153 and the interference portions 157) put in contact with the inner wall surrounding the hollow portion of the core pipe 202 is eight.

FIG. 13 is a diagram describing action of the fitting members according to one embodiment of the present disclosure with parameters such as a distance between the fitting members. In considering the tilting of the flange unit due to the weight of the roll paper, the force illustrated in FIG. 13 acts on each element of the roll paper unit 201. In other words, Mg is the weight of a combination of the core pipe 202 and the rolled up roll paper (the weight of the roll

11

paper portion). F1 is reaction force that acts on each fitting member 136 (a point a) from the inner wall surrounding the hollow portion of the core pipe 202, F2 is reaction force that acts on each fitting member 137 (a point b) from the inner wall surrounding the hollow portion of the core pipe 202, and N is reaction force that acts on the flange sliding shaft portion 122 from the flange bearing 71. Note that, the weight Mg of the roll paper portion is different in accordance with the width of the roll paper used in the printing apparatus and is different also in accordance with the length of the roll paper rolled up around the core pipe 202. The weight Mg of the roll paper portion indicates the maximum value in a case where the roll paper with the maximum width is unused. In the present consideration, the weight Mg of the roll paper portion in this case is used. This is for considering a state where the tilt of the flange unit is the greatest. The point of action of force such as the points a and b is the center of a figure of a contact region (point, line, or surface) that is formed in a case where the fitting member and the sliding axis are put in contact with the core pipe and the bearing. In FIG. 13, A is a distance between the fitting member 136 (the point a) and the fitting member 137 (the point b), and B is a distance between the fitting member 137 (the point b) and the flange bearing 71 (the point of action). Here, those distances A and B are distances along the insertion direction in a case where the flange core portion 132 is inserted into the core pipe 202 of the roll paper portion. Note that, as a matter of course, the following descriptions are given with a calculation result taking into consideration also the flange unit on the non-conveyance-reference side.

The drags F1, F2, and N are obtained as functions of the weight Mg and the distances A and B using the relationship of the forces illustrated in FIG. 13, that is, based on a combination of a force balance of the roll paper portion in the weight direction, a force balance of the flange units on the two sides in the weight direction, and a moment balance at each of the flange units on the two sides. Accordingly, the elastic deformation amount due to the above-described reaction force F1 and F2 at the fitting member 136 (the point a) and the fitting member 137 (the point b) can be obtained, and subsequently, based on this elastic deformation amount, the tilt moduli of the fitting member 136 (the point a) and the fitting member 137 (the point b) with respect to the distance A can be obtained. At last, based on the obtained drags and tilt moduli described above, the tilt modulus of the flange unit on each of the two sides can be obtained.

According to the tilt modulus, it is desirable for the distance A between the first fitting member 136 and the second fitting member 137 to be equal to or greater than the distance B between the second fitting member 137 and the flange bearing 71 (the distance A is equal to or greater than the distance B). This is because the above-described tilt modulus has a characteristic to be smaller in inverse proportion to a distance ratio A/B squared. In other words, regarding this characteristic, it is confirmed in the printing apparatus of the present embodiment that the tilting of the flange unit is effectively suppressed as long as A/B is one or greater, experimentally. On the other hand, for example, the upper limit of A/B can be determined in terms of ease of handling the flange member taking into consideration the lengths of the flange core portions 132 and 134 on which the fitting members are disposed.

Note that, in terms of suppressing the tilting of the flange unit, the form of the fitting members is not limited to the above-described leaf spring form. The fitting members may be any elastic body as long as that the elastic body is disposed on the flange core portion 132, 134 at a predeter-

12

mined distance in the length direction of the flange core portion 132, 134 and is put in contact with the roll paper core pipe 202 with appropriate elasticity while the flange unit is attached thereto. The fitting members may be a rubber member, for example.

In the above-described example, the first and second fitting members 136 and 137 are disposed to be arrayed on the same straight line along the longitudinal direction of the flange core portion 132, 134. In other words, the first and second fitting members 136 and 137 are disposed at the same position in the circumferential direction of the flange core portion 132, 134. However, it is not limited to this form. As long as the first fitting member 136 and the second fitting member 137 are provided away from each other at a predetermined distance along the longitudinal direction as described above, the first fitting member 136 and the second fitting member 137 may be disposed at different positions in the circumferential direction of the flange core portion.

Additionally, the number of the positions at which such elastic bodies are disposed in the circumferential direction of the flange core portion (positions at which the elastic bodies are put in contact with the inner wall surrounding the hollow portion of the core pipe) is two for one fitting member and is four in total for the two fitting members in the above-described embodiment; however, it is not limited thereto. Preferably, the positions at which such elastic bodies are disposed in the circumferential direction of the flange core portion are at least three positions. It is favorable that the three positions are arranged at intervals of 120 degrees. In other words, it is favorable that such elastic bodies are arranged at positions at three equal intervals in the circumferential direction of the flange core portion, and the elastic bodies are put in contact with the inner wall surrounding the hollow portion of the core portion at those positions. Thus, it is possible to align the centers of the flange unit and the hollow portion.

Embodiment 2

FIG. 14 is a perspective view illustrating the flange unit 123 on the conveyance-reference side according to an embodiment 2 of the present disclosure. The present embodiment is different from the above-described embodiment in that the fitting members 136 and 137 on the flange core portion 132 are operated by a cam member 160 that is provided inside the flange core portion 132 to form a mechanism for operating the fitting members 136 and 137. In other words, the cam member 160 pivots between a pressing position in which the fitting members 136 and 137 press the core pipe 202 of the roll paper and a retraction position in which the fitting members 136 and 137 is not put in contact with the core pipe 202 of the roll paper. In this regard, below the fitting members 136, 137, an opening is formed in a portion of the flange core portion 132. Note that, as with the above-described embodiment, since the flange unit 124 on the non-conveyance-reference side has a similar structure, only the flange unit 123 on the conveyance-reference side is described in the descriptions below. It is obvious from the following descriptions that it is possible also in the present embodiment to suppress the tilting of the flange units 123, 124 as with the above-described embodiment.

As illustrated in FIG. 14, inside of the flange core portion 132 is hollow, and a cam member 160 is provided in the hollow so as to be pivotable about the longitudinal direction axis thereof. FIG. 15 illustrates the cam member 160. As illustrated in FIG. 15, the cam member 160 includes two

13

pressing portions 161 and two pressing portions 162. Those pressing portions 161 and 162 correspond to the two fitting members 136 and 137 provided to the flange core portion 132, respectively, and are put in contact with and press the flat surface portions 152 and 156 of the fitting members 136 and 137 in accordance with the pivoting of the cam member 160. Additionally, the cam member 160 includes sliding shaft portions 160A and 160B in the form of a column at two ends thereof, respectively, and those sliding shaft portions 160A and 160B are slidably held by corresponding portions of the flange unit 123 as described later in details with reference to FIG. 16 and the like. Moreover, a lever portion 163 is provided on the circumference of the sliding shaft portion 160B at one end. The user can operate the lever portion 163 so as to move the pressing portions 161 and 162 by pivoting the cam member 160 between the position at which the flat surface portions 152 and 156 of the fitting members 136 and 137 are pressed (the pressing position) and the position at which no pressing is performed (the retraction position). A pivoting range of the lever portion 163 for this movement between the pressing position and the retraction position is indicated by θ in FIG. 15, and the pivoting is restricted within this range by the engagement relationship between the lever portion 163 the flange unit 123 as described with reference to FIG. 16 and the like.

In the present embodiment, by pivoting, the cam member 160 can reciprocate between the pressing position in which the cam member 160 presses the fitting members 136 and 137 and the retraction position in which the cam member 160 retracts from the pressing position. Attaching the flange unit 123 to the roll paper portion while the cam member 160 is moved to the retraction position makes it possible to reduce the resistance between the fitting members 136 and 137 and the inner wall surrounding the hollow portion of the core pipe during the attaching and to improve the operation performance. Here, the retraction position is a position in which the fitting members 136 and 137 are not put in contact with the inner wall surrounding the hollow portion of the core pipe or at which the fitting members 136 and 137 are put in contact with the inner wall but are easily deformed due to the contact (by relatively small force).

In other words, the inner diameter of the hollow portion of the core pipe 202 of the roll paper unit may have a variation depending on the type and the production lot of the roll paper. Here, if the fitting members 136 and 137 are formed in accordance with the dimension of the hollow portion with a small inner diameter in the embodiment 1, the fitting members 136 and 137 cannot interfere and cannot be put in contact with the hollow portion at the time of inserting the fitting members 136 and 137 into the hollow portion with a relatively great inner diameter. For this reason, the roll paper portion cannot be held appropriately, and additionally the tilting of the flange unit 123 described above in the embodiment 1 cannot be suppressed. Therefore, in the embodiment 1, the dimension and the like of the fitting members 136 and 137 are determined in accordance with the dimension of the hollow portion with a great inner diameter. However, in this case, the fitting members 136 and 137 are deformed greatly to be inserted in a case of attaching those fitting members to the hollow portion with a relatively small inner diameter, and the force required for the attachment is increased.

To deal with this, in the present embodiment, the fitting members 136 and 137 are moved to the retraction position at the time of attaching the flange unit 123 to the roll paper portion as described above. On the other hand, in a case where the flange unit 123 is attached to the roll paper portion

14

and thereafter the roll paper unit 201 is formed to be used, the fitting members 136 and 137 are moved to the pressing position in which the fitting members 136 and 137 are put in contact with and press the hollow portion of the roll paper.

Referring back to FIG. 14. In the structure of the flange unit 123, the fitting members 136 and 137 reciprocate between the pressing position and the retraction position by the above-described cam member 160. For this configuration, the flange core portion 132 below the fitting members 136 and 137 is hollow. Thus, the flat surface portions 152 and 156 of the fitting members 136 and 137 and the pressing portions 161 and 162 of the cam member 160 can be put in contact with each other.

The first fitting member 136 is screwed on the conveyance-reference side flange unit 123 by the fixing portion 150, and the second fitting member 137 is screwed on the conveyance-reference side flange unit 123 by the fixing portion 154.

As illustrated in details in FIGS. 16A and 16B, the lever portion 163 of the cam member 160 is positioned on the back surface side of the flange 131, in other words, a side on which the conveyance-reference side flange sliding shaft portion 122 is disposed. The flange 131 has a recess portion formed of a groove extending in the circumferential direction on the back surface side, and the lever portion 163 can be operated in the recess portion.

FIGS. 16A to 16D are diagrams describing the engagement relationship between the cam member 160 and the flange unit 123 and the pressing and the retraction with respect to the fitting members 136 and 137 by the pivoting of the cam member 160. Among the drawings, FIGS. 16A and 16B are cross-sectional views in which the conveyance-reference side flange unit 123 is viewed from a radial direction of the roll paper, and FIGS. 16C and 16D are cross-sectional views in which the conveyance-reference side flange unit 123 is viewed from a rotation axis direction of the roll paper unit 201. FIGS. 16A and 16C illustrate orientations of the fitting members 136 and 137 and the cam member 160 in a position 1, and FIGS. 16B and 16D illustrate orientations of the fitting members 136 and 137 and the cam member 160 in a position 2.

As illustrated in FIGS. 16A and 16B, the sliding shaft portion 160A at one end of the cam member 160 is slidably engaged with a leading end portion of the flange core portion 132, and the sliding shaft portion 160B at the other end is slidably engaged with a groove bottom portion on back surface side of the flange 131. Thus, the cam member 160 is pivotably held by the flange unit 123, and each pivot position is maintained. The lever portion 163 of the cam member 160 is provided from the extending portion of the sliding shaft portion 160B so as to extend perpendicularly to the extending portion. The lever portion 163 extends to the outside of a protrusion portion 131A forming a part of the flange 131 through a slot (groove) formed on the protrusion portion 131A, and thus the user can operate the lever portion 163. This slot is formed on a part of the circumferential direction of the protrusion portion 131A, and thus the pivoting range of the lever portion 163 is restricted to the angle range θ illustrated in FIG. 15.

As illustrated in FIGS. 16A and 16C, the position 1 is the retraction position in which the pressing portions 161 and the pressing portions 162 of the cam member 160 do not press the flat surface portion 152 of the first fitting member 136 and the flat surface portion 156 of the second fitting member 137, respectively. On the other hand, as illustrated in FIGS. 16B and 16D, the position 2 is the pressing position in which the pressing portions 161 and the pressing portions

15

162 press the flat surface portion 152 of the first fitting member 136 and the flat surface portion 156 of the second fitting member 137, respectively. It is possible to switch between the position 1 and the position 2 by operating the lever portion 163 in the direction to pivot with respect to the longitudinal direction axis line of the roll paper unit 201.

FIGS. 17A to 17D are diagrams describing a configuration related to the deformation of the fitting members. Among the drawings, FIGS. 17A and 17B illustrate cross-sectional views along a radial direction of the flange unit, and FIGS. 17C and 17D illustrate cross-sectional views along a rotation axis direction of the flange unit. As described above, the cam member 160 is switched to the position 1 in a case of attaching the flange unit 123 to the roll paper portion. In this position 1, the interference portion 153 of the first fitting member 136 and the interference portion 157 of the second fitting member 137 may be put in contact with the inner wall surrounding the hollow portion of the core pipe 202 of the roll paper portion. With this contact, the first fitting member 136 and the second fitting member 137 are deformed in a direction away from the core pipe 202. In this case, as illustrated in FIG. 6, the interference portions 153 and the fixing portion 150 are apart from each other by a predetermined distance, the interference portions 157 and the fixing portion 154 are apart from each other by a predetermined distance. The width of the constricted portion 151 and the constricted portion 155 is set small. Thus, it is possible to reduce the force required to deform the first fitting member 136 and the second fitting member 137. Thus, it is possible to reduce the frictional force between the interference portions 153, 157 and the inner wall surrounding the hollow portion of the roll paper core pipe 202, and it is possible to reduce the attachment load. As a result, the operation performance at the time of attaching the flange unit 123 to the roll paper portion is improved. After the flange unit 123 is inserted into the roll paper portion, the cam member 160 is switched to the position 2 as illustrated in FIGS. 17B and 17D. As described above, in the position 2, the pressing portions 161 and the pressing portions 162 press the flat surface portion 152 of the first fitting member 136 and the flat surface portion 156 of the second fitting member 137, respectively. As a result, the interference portions 153 and the interference portions 157 are put in contact with the core pipe 202 of the roll paper, and the flange unit 123 is fixed into the core pipe 202. The first fitting member 136 and the second fitting member 137 are elastic bodies. Accordingly, with the elastic deformation of those fitting members 136 and 137, it is possible to suppress the displacement of the roll paper portion in which the flange unit 123 is fixed into the core pipe 202 of the roll paper portion by absorbing the variation of the inner diameter of the core pipe 202 of the roll paper.

Note that, the cam member 160 includes the pressing portions 161 and 162 corresponding to the first and second fitting members 136 and 137 in the above-described embodiment; however, only the pressing portions 161 may be included corresponding to only the first fitting members 136 at the position far from the flange 131. According to this form, a configuration in which the first fitting members 136 that are put in contact with the core pipe 202 first at the time of inserting the flange core portion 132 into the core pipe 202 of the roll paper can be retracted is obtained, and it is possible to secure the operation performance of attaching the flange unit also in this form.

Additionally, the positions 1 and 2 of the fitting members 136 and 137 are switched by operating the lever portion of the cam member in the above-described embodiment 2;

16

however, it is not limited to this form. For example, the force in a longitudinal vertical axis direction of the flange core portion 132, 134 may act through a spring. In this case, a member that develops force in the axis direction by the cam action into a direction crossing a longitudinal vertical axis is used. The fitting members are then pressed by the member.

Embodiment 3

An embodiment 3 of the present disclosure relates to the interference portions 153 and 157 of the first fitting member 136 and the second fitting member 137. FIGS. 18A and 18B are diagrams describing the shape and action of the interference portions of the fitting member. The fitting members 136 and 137 of the conveyance-reference side flange unit 123 are described below, and descriptions of the fitting members 136 and 137 of the non-conveyance-reference side flange unit 124 are omitted since they have similar configurations.

As illustrated in FIG. 18A, a corner portion 158 is provided to each interference portion 153 of the first fitting member 136. As illustrated in FIG. 18B, in the position 2, the flat surface portion 152 of the first fitting member 136 is pressed by the pressing portion 161 of the cam member 160, and the interference portion 153 is put in contact with the inner wall surrounding the hollow portion of the core pipe 202 of the roll paper portion. Thus, in a case where any force acts in a direction in which the roll paper portion is drawn from the conveyance-reference side flange unit 123 (a direction in which the roll paper portion is moved from right to left of FIG. 18B with respect to the conveyance-reference side flange unit 123 in FIG. 18B), the corner portions 158 dig into the wall of the core pipe 202. Thus, it is possible to suppress the movement (displacement) of the roll paper portion with respect to the conveyance-reference side flange unit 123. Note that, in a case of the non-conveyance-reference side, the above-described drawing direction is changed to the opposite direction.

FIG. 19 is a diagram illustrating details of the corner portion. As illustrated in FIG. 19, in a case where an angle formed by one of two crossing straight lines forming the corner portion 158 and the longitudinal direction axis of the core pipe 202 is α , and an angle formed by the other of the two crossing straight lines is β , it is favorable that $\alpha < \beta$. Here, as illustrated in FIGS. 18 and 19, α is an angle formed by the straight line farther from the flange 131 out of the two straight lines forming the corner portion 158, and β is an angle formed by the straight line closer from the flange 131 out of the two straight lines. In other words, α is an angle formed by the straight line farther from an end portion of the core portion on a rear side in the insertion direction into the core pipe, and β is an angle formed by the straight line closer therefrom. Since the two crossing straight lines have such an angle relationship, the corner portion 158 can have a directionality to be against the direction in which the roll paper portion is drawn from the flange core portion 132, and it is possible to suppress the movement (displacement) of the roll paper portion with respect to the conveyance-reference side flange unit 123 more successfully. An apex of the corner portion 158 has a substantially round shape, and a radius R of this rounded portion can be determined in accordance with a desired amount of the digging of the corner portion 158 into the wall of the core pipe 202.

According to the above embodiment, since the function of suppressing the displacement of the roll paper portion with respect to the conveyance-reference side flange unit 123 is added, it is possible to reduce the required pressing amount

17

by the pressing portions **161** to the first fitting member **136**. Consequently, it is possible to further reduce the operation force required to operate the lever portion **163** from the position **1** to the position **2**.

The shape of the corner portion **158** of the first fitting member **136** is described above; however, the second fitting member **137** may have a corner portion shape similar to that of the first fitting member **136** as illustrated in FIG. **20**. Additionally, as illustrated in FIG. **20**, the fixing portion **154** of the second fitting member **137** may have the same configuration as that of the fixing portion **150** of the first fitting member **136**, and a form in which a portion close to the conveyance-reference side flange **131** is pressed by the second fitting member **137** as modified may be applied.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-043691 filed Mar. 18, 2022, which is hereby incorporated by reference wherein in its entirety.

What is claimed is:

1. A printing apparatus that performs printing on a sheet drawn out from a roll in which the sheet is rolled up around a core pipe, comprising:

a roll holding unit including a flange and configured to hold the roll;

a core portion extending from one side of the flange along a rotation axis of the roll and configured to be inserted into a hollow portion of the core pipe from an end portion of the core pipe;

a shaft portion extending from the other side of the flange along the rotation axis;

a flange bearing including a bearing and configured to rotatably support the shaft portion by the bearing in a case where the roll holding unit is mounted on the printing apparatus;

a contact member provided to the core portion and configured to be able to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and

a cam member disposed inside the core portion and configured to rotate about the rotation axis between a first position in which the contact member is pressed onto the inner wall and a second position in which the contact member is retracted from the inner wall.

2. The printing apparatus according to claim 1, wherein the cam member pivots in accordance with an operation of a lever portion attached to the cam member.

3. The printing apparatus according to claim 1, wherein the contact member is in a form of a leaf spring formed by bending a plate member at two bent portions and includes a flat surface portion and two wing-shaped portions continuously contact with the flat surface portion at a tilt through the two bent portions, and the cam member is able to press the flat surface portion.

18

4. The printing apparatus according to claim 3, wherein the two wing-shaped portions each include a corner portion that is put in contact with the inner wall of the core pipe.

5. The printing apparatus according to claim 4, wherein the corner portion of the wing-shaped portion has an angle formed by two crossing straight lines, and while the core portion is inserted in the core pipe, an angle formed by the inner wall and one of the two straight lines that is closer to an end portion of the core portion on a rear side in an insertion direction of the core portion into the core pipe is greater than an angle formed by the inner wall and the other one of the two straight lines.

6. The printing apparatus according to claim 1, wherein the flange is abutted onto an end portion of the core portion.

7. The printing apparatus according to claim 1, wherein the contact member includes a first contact member and a second contact member that is disposed at a position closer to an end portion of the core pipe than the first contact member in a case where the core portion is inserted in the core pipe.

8. The printing apparatus according to claim 7, wherein the cam member is able to pivot to the first position in which the first contact member and the second contact member are pressed onto the inner wall and the second position in which the first contact member and the second contact member are retracted from the inner wall.

9. The printing apparatus according to claim 1, wherein the number of the roll holding unit included is two, and the two roll holding units are inserted into the hollow portion of the core pipe from two ends of the core pipe, respectively.

10. The printing apparatus according to claim 1, wherein the contact member has elasticity.

11. A roll holding device that holds a roll in which a sheet is rolled up around a core pipe, comprising:

a core portion extending from one side of a flange of a roll holding unit along a rotation axis of the roll and configured to be inserted into a hollow portion of the core pipe from an end portion of the core pipe;

a shaft portion extending from the other side of the flange along the rotation axis;

a flange bearing including a bearing and configured to rotatably support the shaft portion by the bearing in a case where the roll holding unit is mounted on the printing apparatus;

a contact member provided to the core portion and configured to be able to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and

a cam member disposed inside the core portion and configured rotate about the rotation axis between a first position in which the contact member is pressed onto the inner wall and a second position in which the contact member is retracted from the inner wall.

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